0732290 0613958 047

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Date: January 26, 1999
To: Purchasers of API 510, *Pressure Vessel Inspection Code: Maintenance Inspection, Rating, Repair, and Alteration*, Eighth Edition
Re: Addendum 1 Errata

This errata contains a correction to Section 6.4, Internal and On-Stream Inspection, on page 6-2. The new text added in the seventh paragraph in this section currently reads:

> If the requirements of item b above are not met, as a result of conditions noted during the scheduled on-stream inspection, the next scheduled inspection shall be an internal inspection. When a vessel has been internally inspected, the results of this inspection can be used to determine whether an on-stream inspection can be substituted for an internal inspection on a similar vessel operating in the same service and conditions.

The correct text should read as follows:

> If the requirements of item b above are not met, as a result of conditions noted during the scheduled on-stream inspection, the next scheduled inspection shall be an internal inspection. When a vessel has been internally inspected, the results of *a current* inspection can be used to determine whether an on-stream inspection can be substituted for an internal inspection on a similar vessel operating in the same service and conditions.

Please replace pages 6-1 and 6–2 with the attached corrected pages.

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STD.API/PETRO API 510-ENGL 1997 ■ 0732290 0613930 T68 ■

6 Inspection and Testing of Pressure Vessels and Pressure-Relieving Devices

6.1 GENERAL

Pressure vessels shall be inspected at the time of installation. Internal field inspections of new vessels are not required as long as a manufacturer's data report assuring that the vessels are satisfactory for their intended service is available. To ensure vessel integrity, all pressure vessels shall be inspected at the frequencies provided in this section.

In selecting the technique(s) to be used for the inspection of a pressure vessel, both the condition of the vessel and the environment in which it operates should be taken into consideration. The inspection, as deemed necessary by the authorized pressure vessel inspector, may include many of a number of nondestructive techniques, including visual inspection. Internal inspection is preferred because process side degradation (corrosion, erosion, and environmental cracking) can be nonuniform throughout the vessel and, therefore, difficult to locate by external NDE. On-stream inspection may be acceptable in lieu of internal inspection for vessels under the specific circumstances defined in 6.4. In situations where on-stream inspection is acceptable, such inspection may be conducted either while the vessel is out of service and depressurized or on stream and under pressure. Except in response to an apparent need, such as when environmental cracking (see *Guide for Inspection of Refinery Equipment,* Chapter II) is suspected, inspection techniques exceeding the examination requirements used in the design and fabrication of the vessel are not required.

The appropriate inspection must provide the information necessary to determine that all of the essential sections or components of the vessel are safe to operate until the next scheduled inspection. The risks associated with operational shutdown and start-up and the possibility of increased corrosion due to exposure of vessel surfaces to air and moisture should be evaluated when an internal inspection is being planned.

6.2 RISK-BASED INSPECTION

Identifying and evaluating potential degradation mechanisms are important steps in an assessment of the likelihood of a pressure vessel failure. However, adjustments to inspection strategy and tactics to account for consequences of a failure should also be considered. Combining the assessment of likelihood of failure and the consequence of failure are essential elements of risk-based inspection (RBI).

When an owner/user chooses to conduct a RBI assessment, it must include a systematic evaluation of both the likelihood of failure and the associated consequence of failure. The likelihood assessment must be based on all forms of degradation that could reasonably be expected to affect a vessel in any particular service. Examples of those degradation mechanisms include: internal or external metal loss from an identified form of corrosion (localized or general), all forms of cracking, including hydrogen assisted and stress corrosion cracking (from the inside or outside surfaces of a vessel), and any other forms of metallurgical, corrosion, or mechanical degradation, such as fatigue, embrittlement, creep, etc. Additionally, the effectiveness of the inspection practices, tools, and techniques utilized for finding the expected and potential degradation mechanisms must be evaluated. This likelihood of failure assessment should be repeated each time equipment or process changes are made that could significantly affect degradation rates or cause premature failure of the vessel.

Other factors that should be considered in a RBI assessment include: appropriateness of the materials of construction; vessel design conditions, relative to operating conditions; appropriateness of the design codes and standards utilized; effectiveness of corrosion monitoring programs; and the quality of maintenance and inspection quality assurance/quality control programs. Equipment failure data and information will also be important information for this assessment. The consequence assessment must consider the potential incidents that may occur as a result of fluid release, including explosion, fire, toxic exposure, environmental impact, and other health effects associated with a failure of a vessel.

It is essential that all RBI assessments be thoroughly documented, clearly defining all the factors contributing to both the likelihood and consequence of a failure of the vessel.

After an effective RBI assessment is conducted, the results can be used to establish a vessel inspection strategy and more specifically better define the following:

a. The most appropriate inspection methods, scope, tools and techniques to be utilized based on the expected forms of degradation.

b. The appropriate frequency for internal, external, and on-stream inspections.

c. The need for pressure testing after damage has been incurred or after repairs or modifications have been completed.

d. The prevention and mitigation steps to reduce the likelihood and consequence of a vessel failure.

An RBI assessment may be used to increase or decrease the 10-year inspection limit described in Section 6.4. When used to increase the 10-year limit, RBI assessment shall be reviewed and approved by a pressure vessel engineer and authorized pressure vessel inspector at intervals not to exceed 10 years, or more often if warranted by process, equipment, or consequence changes.

6.3 EXTERNAL INSPECTION

Each vessel aboveground shall be given a visual external inspection, preferably while in operation, at least every 5 years or at the same interval as the required internal or on-stream inspection, whichever is less. The inspection shall, at the least,

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determine the condition of the exterior insulation, the condition of the supports, the allowance for expansion, and the general alignment of the vessel on its supports. Any signs of leakage should be investigated so that the sources can be established. Inspection for corrosion under insulation (CUI) shall be considered for externally-insulated vessels subject to moisture ingress and that operate between 25°F (–4°C) and 250°F (120°C), or are in intermittent service. This inspection may require removal of some insulation. It is not normally necessary to remove insulation if the entire vessel shell is always operated at a temperature sufficiently low [below 25°F (–4°C)] or sufficiently high [above 250°F (120°C)] to prevent the presence or condensation of moisture under the insulation. Alternatively, shell thickness measurements done internally at typical problem areas (for example, stiffening rings, around nozzles, and other locations which tend to trap moisture or allow moisture ingress) may be performed during internal inspections.

Buried vessels shall be inspected to determine their external environmental condition. The inspection interval shall be based on corrosion-rate information obtained from one or more of the following methods: (a) during maintenance activity on adjacent connecting piping of similar material; (b) from the interval examination (specified in the paragraph above) of similarly buried corrosion test coupons of similar material; (c) from representative portions of the actual vessel; or (d) from a vessel in similar circumstances.

Vessels that are known to have a remaining life of over 10 years or that are protected against external corrosion—for example, (a) vessels insulated effectively to preclude the entrance of moisture, (b) jacketed cryogenic vessels, (c) vessels installed in a cold box in which the atmosphere is purged with an inert gas, and (d) vessels in which the temperature being maintained is sufficiently low or sufficiently high to preclude the presence of water—do not need to have insulation removed for the external inspection. However, the condition of their insulating system or their outer jacketing, such as the cold box shell, shall be observed at least every 5 years and repaired if necessary.

6.4 INTERNAL AND ON-STREAM INSPECTION

The period between internal or on-stream inspections shall not exceed one half the estimated remaining life of the vessel based on corrosion rate or 10 years, whichever is less. In cases where the remaining safe operating life is estimated to be less than 4 years, the inspection interval may be the full remaining safe operating life up to a maximum of 2 years.

For pressure vessels that are in noncontinuous service and are isolated from the process fluids such that they are not exposed to corrosive environments (such as inert gas purged or filled with noncorrosive hydrocarbons), the 10 years shall be the 10 years of actual service exposed life. Equipment that is not adequately protected from corrosive environments may experience significant internal corrosion while idle and should be carefully reviewed when setting inspection intervals. In no case should these exceed one-half the estimated remaining corrosion-rate life, or 10 years since the last inspection. External inspections for vessels in noncontinuous service remain the same as for continuous service, as outlined in 6.3.

Except as noted below, internal inspection is normally the preferred method of inspection and shall be conducted on vessels subject to significant localized corrosion and other types of damage. At the discretion of the authorized pressure vessel inspector, on-stream inspection may be substituted for internal inspection in the following situations:

a. When size, configuration, or lack of access makes vessel entry for internal inspection physically impossible.

b. When the general corrosion rate of a vessel is known to be less than 0.005 inch (0.125 millimeter) per year and the estimated remaining life is greater than 10 years, and all of the following conditions are met:

1. The corrosive character of the contents, including the effect of trace components, has been established by at least 5 years of the same or comparable service experience with the type of contents being handled.
2. No questionable condition is disclosed by the external inspection specified in 6.3.
3. The operating temperature of the steel vessel shell does not exceed the lower temperature limits for the creep-rupture range of the vessel material.
4. The vessel is not considered to be subject to environmental cracking or hydrogen damage from the fluid being handled.
5. The vessel is not strip-lined or plate-lined.

If the requirements of item b above are not met, as a result of conditions noted during the scheduled on-stream inspection, the next scheduled inspection shall be an internal inspection. When a vessel has been internally inspected, the results of a current inspection can be used to determine whether an on-stream inspection can be substituted for an internal inspection on a similar vessel operating in the same service and conditions.

When an on-stream inspection is conducted in lieu of an internal inspection, a thorough examination shall be performed using ultrasonic thickness measurements, or radiography, or other appropriate means of NDE to measure metal thicknesses and/or assess the integrity of the metal and welds. If an on-stream inspection is conducted, the authorized pressure vessel inspector shall be given sufficient access to all parts of the vessel (heads, shell, and nozzles) so that the inspector is satisfied that an accurate assessment of the vessel condition can be made.

A representative number of thickness measurements must be conducted on each vessel to satisfy the requirements for an internal or on-stream inspection. For example, the thickness for all major components (shells, heads, cone sections) and a rep-

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Date: December 1998
To: Purchasers of API 510, *Pressure Vessel Inspection Code: Maintenance Inspection, Rating, Repair, and Alteration*, Eighth Edition
Re: Addendum 1

This package contains Addendum 1 of API 510, *Pressure Vessel Inspection Code: Maintenance Inspection, Rating, Repair, and Alteration,* Eighth Edition. This package consists of the pages that have changed since the June 1997 printing of API 510.

To update your copy of API 510, replace the following pages as indicated:

Part of Book Changed	Old Pages to be Replaced	New Pages
Cover	front covers	front cover and inside front cover
	back cover	back cover and inside back cover
Title Page	title page and Special Notes page	title page and Special Notes page
Foreword	iii–vi	iii–v (+ blank)
Table of Contents	vii–viii	vii–viii
Section 2	2-1	2-1 (+blank)
Section 3	3-1–3-2	3-1–3-2
Section 4	4-1	4-1 (+blank)
Section 5	5-1–5-4	5-1–5-4
Section 6	6-1–6-4	6-1–6-5 (+blank)
Section 8	8-3	8-3 (+blank)
Appendix B	B-1	B-1 (+blank)

The parts of the text, tables, and figures that contain changes are indicated by a vertical bar and a small "98" in the margin.

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Pressure Vessel Inspection Code: Maintenance Inspection, Rating, Repair, and Alteration

API 510
EIGHTH EDITION, JUNE 1997
ADDENDUM 1, DECEMBER 1998





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- To make safety, health and environmental considerations a priority in our planning, and our development of new products and processes.
- To advise promptly, appropriate officials, employees, customers and the public of information on significant industry-related safety, health and environmental hazards, and to recommend protective measures.
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- To work with others to resolve problems created by handling and disposal of hazardous substances from our operations.
- To participate with government and others in creating responsible laws, regulations and standards to safeguard the community, workplace and environment.
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Pressure Vessel Inspection Code: Maintenance Inspection, Rating, Repair, and Alteration

Manufacturing, Distribution and Marketing Department

API 510
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FOREWORD

In December 1931, API and the American Society of Mechanical Engineers (ASME) created the Joint API/ASME Committee on Unfired Pressure Vessels. This committee was created to formulate and prepare for publication a code for safe practices in the design, construction, inspection, and repair of pressure vessels to be used in the petroleum industry. Entitled API/ASME *Code for Unfired Pressure Vessels for Petroleum Liquids and Gases* (commonly called the API/ASME *Code for Unfired Pressure Vessels* or API/ASME Code), the first edition of the code was approved for publication in 1934.

From its inception, the API/ASME Code contained Section I, which covered recommended practices for vessel inspection and repair and for establishing allowable working pressures for vessels in service. Section I recognized and afforded well-founded bases for handling various problems associated with the inspection and rating of vessels subject to corrosion. Although the provisions of Section I (like other parts of the API/ASME Code) were originally intended for pressure vessels installed in the plants of the petroleum industry, especially those vessels containing petroleum gases and liquids, these provisions were actually considered to be applicable to pressure vessels in most services. ASME's Boiler and Pressure Vessel Committee adopted substantially identical provisions and published them as a nonmandatory appendix in the 1950, 1952, 1956, and 1959 editions of Section VIII of the ASME *Boiler and Pressure Vessel Code.*

After the API/ASME Code was discontinued in 1956, a demand arose for the issuance of Section I as a separate publication, applicable not only to vessels built in accordance with any edition of the API/ASME Code but also to vessels built in accordance with any edition of Section VIII of the ASME Code. Such a publication appeared to be necessary to assure industry that the trend toward uniform maintenance and inspection practices afforded by Section I of the API/ASME Code would be preserved. API 510, first published in 1958, is intended to satisfy this need.

The procedures in Section I of the 1951 edition of the API/ASME Code, as amended by the March 16, 1954 addenda, have been updated and revised in API 510. Section I of the API/ASME Code contained references to certain design or construction provisions, so these references have been changed to refer to provisions in the ASME Code. Since the release of the 1960 edition of the *National Board Inspection Code*, elements of the API/ASME Code have also been carried by the *National Board Inspection Code.*

It is the intent of API to keep this publication up to date. All pressure vessel owners and operators are invited to report their experiences in the inspection and repair of pressure vessels whenever such experiences may suggest a need for revising or expanding the practices set forth in API 510.

98 This edition of API 510 supersedes all previous editions of API 510, *Pressure Vessel Inspection Code: Maintenance Inspection, Rating, and Repair of Pressure Vessels.* Each edition, revision, or addenda to this API standard may be used beginning with the date of issuance shown on the cover page for that edition, revision, or addenda. Each edition, revision, or addenda to this API standard becomes effective 6 months after the date of issuance for equipment that is rerated, reconstructed, relocated, repaired, modified (altered), inspected, and tested per this standard. During the 6-month time between the date of issuance of the edition, revision, or addenda and the effective date, the user shall specify to which edition, revision, or addenda, and the equipment is to be rerated, reconstructed, relocated, repaired, modified (altered), inspected and tested.

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Suggested revisions are invited and should be submitted to the director of the Manufacturing, Distribution and Marketing Department, American Petroleum Institute, 1220 L Street, N.W., Washington, D.C. 20005-4070.

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IMPORTANT INFORMATION CONCERNING USE OF ASBESTOS OR ALTERNATIVE MATERIALS

Asbestos is specified or referenced for certain components of the equipment described in some API standards. It has been of extreme usefulness in minimizing fire hazards associated with petroleum processing. It has also been a universal sealing material, compatible with most refining fluid services.

Certain serious adverse health effects are associated with asbestos, among them the serious and often fatal diseases of lung cancer, asbestosis, and mesothelioma (a cancer of the chest and abdominal linings). The degree of exposure to asbestos varies with the product and the work practices involved.

Consult the most recent edition of the Occupational Safety and Health Administration (OSHA), U.S. Department of Labor, Occupational Safety and Health Standard for Asbestos, Tremolite, Anthophyllite, and Actinolite, 29 *Code of Federal Regulations* Section 1910.1001; the U.S. Environmental Protection Agency, National Emission Standard for Asbestos, 40 *Code of Federal Regulations* Sections 61.140 through 61.156; and the U.S. Environmental Protection Agency (EPA) rule on labeling requirements and phased banning of asbestos products, published at 54 *Federal Register* 29460 (July 12, 1989).

There are currently in use and under development a number of substitute materials to replace asbestos in certain applications. Manufacturers and users are encouraged to develop and use effective substitute materials that can meet the specifications for, and operating requirements of, the equipment to which they would apply.

SAFETY AND HEALTH INFORMATION WITH RESPECT TO PARTICULAR PRODUCTS OR MATERIALS CAN BE OBTAINED FROM THE EMPLOYER, THE MANUFACTURER OR SUPPLIER OF THAT PRODUCT OR MATERIAL, OR THE MATERIAL SAFETY DATA SHEET.

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2 References

The most recent editions of the following standards, codes, and specifications are cited in this inspection code.

API

RP 572 *Inspection of Pressure Vessels*

98 RP 574 *Inspection of Piping System Components*

RP 576 *Inspection of Pressure-Relieving Devices*

98 *API 510 Inspector Certification Examination Body of Knowledge*

Guide for Inspection of Refinery Equipment, Chapter II, "Conditions Causing Deterioration or Failures"

Note: This publication is out of print. To obtain a copy please inform the person taking your order that you require this publication for the API 510 Inspector Certification Exam.

ASME[1]

Boiler and Pressure Vessel Code, Section V, Section VI, Section VII, Section VIII, Section IX, and Section XI

National Board[2]

National Board Inspection Code

[1]ASME International, Three Park Avenue, New York, NY 10016-5990.
[2]National Board of Boiler and Pressure Vessel Inspectors, 1055 Crupper Avenue, Columbus, Ohio 43229.

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3 Definitions

For the purposes of this standard, the following definitions apply.

3.1 alteration: A physical change in any component or a rerating that has design implications that affect the pressure-containing capability of a pressure vessel beyond the scope of the items described in existing data reports. The following should not be considered alterations: any comparable or duplicate replacement, the addition of any reinforced nozzle less than or equal to the size of existing reinforced nozzles, and the addition of nozzles not requiring reinforcement.

3.2 ASME Code: Abbreviation and shortened title for the ASME *Boiler and Pressure Vessel Code*. This abbreviated title includes the addenda and code cases of the ASME *Boiler and Pressure Vessel Code*.

The ASME Code is written for new construction; however, most of the technical requirements for design, welding, examination, and materials can be applied in the maintenance inspection, rating, repair, and alteration of operating pressure vessels. When the ASME Code cannot be followed because of its new construction orientation (new or revised material specifications, inspection requirements, certain heat treatments and pressure tests, and stamping and inspection requirements), the engineer or inspector shall conform to this inspection code rather than to the ASME Code. If an item is covered by requirements in the ASME Code and this inspection code or if there is a conflict between the two codes, for vessels that have been placed in service, the requirements of this inspection code shall take precedence over the ASME Code. As an example of the intent of this inspection code, the phrase "applicable requirements of the ASME Code" has been used in this inspection code instead of the phrase "in accordance with the ASME Code."

3.3 authorized pressure vessel inspector: An employee of an authorized inspection agency who is qualified and certified to perform inspections under this inspection code.

3.4 authorized inspection agency: Any one of the following:

a. The inspection organization of the jurisdiction in which the pressure vessel is used.
b. The inspection organization of an insurance company that is licensed or registered to write and actually does write pressure vessel insurance.
c. The inspection organization of an owner or user of pressure vessels who maintains an inspection organization for his equipment only and not for vessels intended for sale or resale.
d. An independent organization or individual that is under contract to and under the direction of an owner-user and that is recognized or otherwise not prohibited by the jurisdiction in which the pressure vessel is used. The owner-user's inspection program shall provide the controls that are necessary when contract inspectors are used.

3.5 construction code: The code or standard to which a vessel was originally built, such as API/ASME, API, or State Special/non-ASME.

3.6 inspection code: Shortened title for API 510 used in this publication.

3.7 inspector: Refers to an authorized pressure vessel inspector in this document. 98

3.8 jurisdiction: A legally constituted government administration that may adopt rules relating to pressure vessels.

3.9 maximum allowable working pressure: The maximum gauge pressure permitted at the top of a pressure vessel in its operating position for a designated temperature. This pressure is based on calculations using the minimum (or average pitted) thickness for all critical vessel elements, exclusive of thickness designated for corrosion and loadings other than pressure.

3.10 minimum allowable shell thickness: The thickness required for each element of a vessel. The minimum allowable shell thickness is based on calculations that consider temperature, pressure, and all loadings.

3.11 on-stream inspection: The inspection used to establish the suitability of a pressure vessel for continued operation. Nondestructive examination (NDE) procedures are used to establish the suitability of the vessel, and the vessel may or may not be in operation while the inspection is being carried out. Because a vessel may be in operation while an on-stream inspection is being carried out, an on-stream inspection means essentially that the vessel is not entered for internal inspection.

3.12 pressure vessel: A container designed to withstand internal or external pressure. This pressure may be imposed by an external source, by the application of heat from a direct or indirect source, or by any combination thereof. This definition includes unfired steam generators and other vapor generating vessels which use heat from the operation of a processing system or other indirect heat source. (Specific limits and exemptions of equipment covered by this inspection code are given in Section 1 and Appendix A.)

3.13 pressure vessel engineer: Shall be one or more persons or organizations acceptable to the owner-user who are knowledgeable and experienced in the engineering disciplines associated with evaluating mechanical and material characteristics which affect the integrity and reliability of pressure vessels. The pressure vessel engineer, by consult-

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ing with appropriate specialists, should be regarded as a composite of all entities needed to properly assess the technical requirements.

3.14 quality assurance: All planned, systematic, and preventative actions required to determine if materials, equipment, or services will meet specified requirements so that equipment will perform satisfactorily in service. The contents of a quality assurance inspection manual are outlined in 4.3.

3.15 repair: The work necessary to restore a vessel to a condition suitable for safe operation at the design conditions. If any repair changes the design temperature or pressure, the requirements for rerating shall be satisfied. A repair can be the addition or replacement of pressure or nonpressure parts that do not change the rating of the vessel.

3.16 repair organization: Any one of the following:

a. The holder of a valid ASME Certificate of Authorization that authorizes the use of an appropriate ASME Code symbol stamp.
b. An owner or user of pressure vessels who repairs his or her own equipment in accordance with this inspection code.
c. A contractor whose qualifications are acceptable to the pressure-vessel owner or user and who makes repairs in accordance with this inspection code.
d. An individual or organization that is authorized by the legal jurisdiction.

3.17 rerating: A change in either the temperature ratings or the maximum allowable working pressure rating of a vessel, or a change in both. The maximum allowable working temperature and pressure of a vessel may be increased or decreased because of a rerating, and sometimes a rerating requires a combination of changes. Derating below original design conditions is a permissible way to provide for corrosion. When a rerating is conducted in which the maximum allowable working pressure or temperature is increased or the minimum temperature is decreased so that additional mechanical tests are required, it shall be considered an alteration.

3.18 examiner: A person who assists the API authorized pressure vessel inspector by performing specific NDE on pressure vessels but does not evaluate the results of those examinations in accordance with API 510, unless specifically trained and authorized to do so by the owner or user. The examiner need not be certified in accordance with API 510 or be an employee of the owner or user but shall be trained and competent in the applicable procedures in which the examiner is involved. In some cases, the examiner may be required to hold other certifications as necessary to satisfy the owner or user requirements. Examples of other certification that may be required are American Society for Nondestructive Testing[3] SNT-TC-1A, or CP189, or American Welding Society[4] Welding Inspector Certification. The examiner's employer shall maintain certification records of the examiners employed, including dates and results of personnel qualifications and shall make them available to the API authorized pressure vessel inspector.

[3]American Society for Nondestructive Testing, Inc., 1711 Arlingate Lane, P.O. Box 28518, Columbus, Ohio, 43228-0518.
[4]American Welding Society, 550 N.W. LeJeune Road, Miami, FL 33135.

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4 Owner-User Inspection Organization

4.1 GENERAL

An owner-user of pressure equipment shall exercise control of the pressure vessel inspection program, inspection frequencies, and maintenance. The owner-user is responsible for the function of an authorized inspection agency in accordance with the provisions of API 510. The owner-user inspection organization shall control activities relating to the maintenance inspection, rating, repair, and alteration of these pressure vessels. 98

4.2 API AUTHORIZED PRESSURE VESSEL INSPECTOR QUALIFICATION AND CERTIFICATION

Authorized pressure vessel inspectors shall have education and experience in accordance with Appendix B of this inspection code. Authorized pressure vessel inspectors shall be certified by the American Petroleum Institute in accordance with the provisions of Appendix B. 98

4.3 OWNER-USER ORGANIZATION RESPONSIBILITIES

An owner-user organization is responsible for developing, documenting, implementing, executing, and assessing pressure vessel inspection systems and inspection procedures that will meet the requirements of this inspection code. These systems and procedures will be contained in a quality assurance inspection manual and shall include the following:

a. Organization and reports of structure for inspection personnel.
b. Documentation and maintenance of inspection and quality assurance procedures.
c. Documentation and reports of inspection and test results.
d. Corrective action for inspection and test results.
e. Internal audits for compliance with the quality assurance inspection manual.
f. Review and approval of drawings, design calculations, and specifications for repairs, alterations, and reratings.
g. Assurance that all jurisdictional requirements for pressure vessel inspection, repairs, alterations, and rerating are continuously met.
h. Reports to the authorized pressure vessel inspector any process changes that could affect pressure vessel integrity.
i. Training requirements for inspection personnel regarding inspection tools, techniques, and technical knowledge base.
j. Controls necessary so that only qualified welders and procedures are used for all repairs and alterations.
k. Controls necessary so that only qualified nondestructive examination (NDE) personnel and procedures are utilized.
l. Controls necessary so that only materials conforming to the applicable section of the ASME Code are utilized for repairs and alterations.
m. Controls necessary so that all inspection measurement and test equipment are properly maintained and calibrated.
n. Controls necessary so that the work of contract inspection or repair organizations meet the same inspection requirements as the owner-user organization.
o. Internal auditing requirements for the quality control system for pressure-relieving devices.

4.4 API AUTHORIZED PRESSURE VESSEL INSPECTOR RESPONSIBILITIES 98

When inspections, repairs, or alterations are being conducted on pressure vessels, an API authorized pressure vessel inspector shall be responsible to the owner-user for determining that the requirements of API 510 on inspection, examination, and testing are met, and shall be directly involved in the inspection activities. The API authorized pressure vessel inspector may be assisted in performing visual inspections by other properly trained and qualified individuals, who may or may not be certified vessel inspectors. Personnel performing nondestructive examinations shall meet the qualifications identified in 3.17 but need not be API authorized pressure vessel inspectors. However, all examination results must be evaluated and accepted by the API authorized pressure vessel inspector.

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5 Inspection Practices

5.1 PREPARATORY WORK

Safety precautions are important in pressure vessel inspection because of the limited access to and the confined spaces of pressure vessels. Occupational Safety and Health Administration (OSHA) regulations pertaining to confined spaces and any other OSHA safety rules should be reviewed and followed, where applicable.

For an internal inspection, the vessel should be isolated by blinds or other positive methods from all sources of liquids, gases, or vapors. The vessel should be drained, purged, cleaned, ventilated, and gas tested before it is entered. Where required, protective equipment should be worn that will protect the eyes, lungs, and other parts of the body from specific hazards that may exist in the vessel.

The nondestructive testing equipment used for the inspection is subject to the safety requirements customarily followed in a gaseous atmosphere. Before the inspection is started, all persons working around the vessel should be informed that people are going to be working inside it. People working inside the vessel should be informed when any work is going to be done on the exterior of it.

The tools and personnel safety equipment needed for the vessel inspection should be checked before the inspection. Other equipment that might be needed for the inspection, such as planking, scaffolding, bosun's chairs, and portable ladders, should be available if needed.

5.2 MODES OF DETERIORATION AND FAILURE

98 Contaminants in fluids handled in pressure vessels, such as sulfur, chlorine, hydrogen sulfide, hydrogen, carbon, cyanides, acids, water, or other corroding species may react with metals and cause corrosion. Significant stress fluctuations or reversals in parts of equipment are common, particularly at points of high secondary stress. If stresses are high and reversals are frequent, failure of parts may occur because of fatigue. Fatigue failures in pressure vessels may also occur because of cyclic temperature and pressure changes. Locations where metals with different thermal coefficients of expansion are welded together may be susceptible to thermal fatigue.

Deterioration or creep may occur if equipment is subjected to temperatures above those for which it is designed. Since metals weaken at higher temperatures, such deterioration may cause failures, particularly at points of stress concentration. Creep is dependent on time, temperature, stress, and material creep strength, so the actual or estimated levels of these quantities should be used in any evaluations. At elevated temperatures, other metallurgical changes may also take place that may permanently affect equipment.

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For developing an inspection plan for equipment operating at elevated temperatures [generally starting in the range of 750°–1000°F (400°–540°C), depending on operating conditions and alloy], the following should be considered in assessing the remaining life:

a. Creep deformation and stress rupture.
b. Creep crack growth.
c. Effect of hydrogen on creep.
d. Interaction of creep and fatigue.
e. Possible metallurgical effects, including a reduction in ductility.

Numerous NDE techniques can be applied to find and characterize elevated temperature damage. These techniques include visual, surface, and volumetric examination. Additionally, if desired or warranted, samples can be removed for laboratory analysis. 98

The inspection plan should be prepared in consultation with an engineer having knowledge of elevated temperature and metallurgical effects on pressure vessel materials of construction.

At subfreezing temperatures, water and some chemicals handled in pressure vessels may freeze and cause failure.

At ambient temperatures, carbon, low-alloy, and other ferritic steels may be susceptible to brittle failure. A number of failures have been attributed to brittle fracture of steels that were exposed to temperatures below their transition temperature and to pressures greater than 20 percent of the required hydrostatic test pressure; most brittle fractures, however, have occurred on the first application of a particular stress level (the first hydrotest or overload). Although the potential for a brittle failure because of excessive operating conditions below the transition temperature shall be evaluated, the potential for a brittle failure because of rehydrotesting or pneumatic testing of equipment or the addition of any other additional loadings shall also be evaluated. Special attention should be given to low-alloy steels (especially 2¼ Cr-1Mo) because they may be prone to temper embrittlement. [*Temper embrittlement* is a loss of ductility and notch toughness due to postweld heat treatment or high-temperature service (above 700°F) (370°C).]

Other forms of deterioration, such as stress corrosion cracking, hydrogen attack, carburization, graphitization, and erosion, may also occur under special circumstances. These forms of deterioration are more fully discussed in Chapter II of the API *Guide for Inspection for Refinery Equipment.*

5.3 CORROSION RATE DETERMINATION

For a new vessel or for a vessel for which service conditions are being changed, one of the following methods shall be employed to determine the vessel's probable corrosion rate. The remaining wall thickness at the time of the next inspection can be estimated from this rate.

a. A corrosion rate may be calculated from data collected by the owner or user on vessels providing the same or similar service.

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b. If data on vessels providing the same or similar service are not available, a corrosion rate may be estimated from the owner's or user's experience or from published data on vessels providing comparable service.

c. If the probable corrosion rate cannot be determined by either item a or item b above, on-stream determinations shall be made after approximately 1000 hours of service by using suitable corrosion monitoring devices or actual nondestructive thickness measurements of the vessel or system. Subsequent determinations shall be made after appropriate intervals until the corrosion rate is established.

If it is determined that an inaccurate corrosion rate has been assumed, the rate to be used for the next period shall be increased or may be decreased to agree with the actual rate.

5.4 MAXIMUM ALLOWABLE WORKING PRESSURE DETERMINATION

The maximum allowable working pressure for the continued use of a pressure vessel shall be based on computations that are determined using the latest edition of the ASME Code or the construction code to which the vessel was built. The resulting maximum allowable working pressure from these computations shall not be greater than the original maximum allowable working pressure unless a rerating is performed in accordance with 7.3.

Computations may be made only if the following essential details comply with the applicable requirements of the code being used: head, shell, and nozzle reinforcement designs; material specifications; allowable stresses; weld efficiencies; inspection acceptance criteria; and cyclical service requirements. In corrosive service, the wall thickness used in these computations shall be the actual thickness as determined by inspection (see 5.7) but shall not be greater than the thickness reported in the material test report or the manufacturer's data report, where available, minus twice the estimated corrosion loss before the date of the next inspection, except as modified in 6.4. Allowance shall be made for the other loadings in accordance with the applicable provisions of the ASME Code.

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5.5 DEFECT INSPECTION

Vessels shall be examined for visual indications of distortion. If any distortion of a vessel is suspected or observed, the overall dimensions of the vessel shall be checked to confirm whether or not the vessel is distorted and, if it is distorted, to determine the extent and seriousness of the distortion. The parts of the vessel that should be inspected most carefully depend on the type of vessel and its operating conditions. The authorized pressure vessel inspector should be familiar with the operating conditions of the vessel and with the causes and characteristics of potential defects and deterioration. (For recommended inspection practices for pressure vessels, see API Recommended Practice 572.)

Careful visual examination is the most important and the most universally accepted method of inspection. Other methods that may be used to supplement visual inspection include (a) magnetic-particle examination for cracks and other elongated discontinuities in magnetic materials; (b) fluorescent or dye-penetrant examination for disclosing cracks, porosity, or pin holes that extend to the surface of the material and for outlining other surface imperfections, especially in nonmagnetic materials; (c) radiographic examination; (d) ultrasonic thickness measurement and flaw detection; (e) eddy current examination; (f) metallographic examination; (g) acoustic emission testing; hammer testing while not under pressure; and (h) pressure testing. (Section V of the ASME Code can be used as a guide for many of the nondestructive examination techniques.)

Adequate surface preparation is important for proper visual examination and for the satisfactory application of any auxiliary procedures, such as those mentioned above. The type of surface preparations required depends on the individual circumstances, but surface preparations such as wire brushing, blasting, chipping, grinding, or a combination of these preparations may be required.

If external or internal coverings, such as insulation, refractory protective linings, and corrosion-resistant linings, are in good condition and there is no reason to suspect that an unsafe condition is behind them, it is not necessary to remove them for inspection of the vessel; however, it may be advisable to remove small portions of the coverings to investigate their condition and effectiveness and the condition of the metal underneath them.

Where operating deposits, such as coke, are normally permitted to remain on a vessel surface, it is particularly important to determine whether such deposits adequately protect the vessel surface from deterioration. To determine this, spot examinations in which the deposit is thoroughly removed from selected critical areas may be required.

Where vessels are equipped with removable internals, the internals need not be removed completely as long as reasonable assurance exists that deterioration in regions rendered inaccessible by the internals is not occurring to an extent beyond that found in more accessible parts of the vessel.

5.6 INSPECTION OF PARTS

The following inspections are not all inclusive for every vessel, but they do include the features that are common to most vessels and that are most important. Authorized pressure vessel inspectors must supplement this list with any additional items necessary for the particular vessel or vessels involved.

a. Examine the surfaces of shells and heads carefully for possible cracks, blisters, bulges, and other signs of deterioration. Pay particular attention to the skirt and to support-attachment

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and knuckle regions of the heads. If evidence of distortion is found, it may be necessary to make a detailed check of the actual contours or principal dimensions of the vessel and to compare those contours and dimensions with the original design details.

b. Examine welded joints and the adjacent heat-affected zones for service-induced cracks or other defects. On riveted vessels, examine rivet head, butt strap, plate, and caulked edge conditions. If rivet-shank corrosion is suspected, hammer testing or spot radiography at an angle to the shank axis may be useful.

c. Examine the surfaces of all manways, nozzles, and other openings for distortion, cracks, and other defects, paying particular attention to the welding used to attach the parts and their reinforcements. Normally, weep holes in reinforcing plates should remain open to provide visual evidence of leakage as well as to prevent pressure build-up in the cavity. Examine accessible flange faces for distortion and determine the condition of gasket-seating surfaces.

98 | API Recommended Practice 574 provides more information on the inspection of piping, valves, and fittings associated with pressure vessels. API Recommended Practice 572
98 | provides more information on pressure vessel inspection.

5.7 CORROSION AND MINIMUM THICKNESS EVALUATION

Corrosion may cause a uniform loss (a general, relatively even wastage of a surface area) or may cause a pitted appearance (an obvious, irregular surface wastage). Uniform corrosion may be difficult to detect visually, and thickness readings may be necessary to determine its extent. Pitted surfaces may be thinner than they appear visually, and when there is uncertainty about the original surface location, thickness determinations may also be necessary.

The minimum actual thickness and maximum corrosion rate for any part of a vessel may be adjusted at any inspection. When the minimum actual thickness or maximum corrosion rate is to be adjusted, one of the following should be considered:

a. Any suitable nondestructive examination, such as ultrasonic or radiographic examination, that will not affect the safety of the vessel may be used as long as it will provide minimum thickness determinations. When a measurement method produces considerable uncertainty, test holes may be drilled, or other nondestructive techniques, such as ultrasonic A-scan, B-scan, or C-scan, may be employed. Profile radiography may be also employed.

b. If suitable openings are available, measurements may be taken through them.

c. The depth of corrosion may be determined by gauging the uncorroded surfaces within the vessel when such surfaces are in the vicinity of the corroded area.

d. For a corroded area of considerable size in which the circumferential stresses govern, the least thickness along the most critical element of the area may be averaged over a length not exceeding the following:

1. For vessels with inside diameters less than or equal to 60 inches (150 centimeters), one-half the vessel diameter or 20 inches (50 centimeters), whichever is less.

2. For vessels with inside diameters greater than 60 inches (150 centimeters), one-third the vessel diameter or 40 inches (100 centimeters), whichever is less.

When the area contains an opening, the distance on either side of the opening within which the thicknesses are averaged shall not extend beyond the limits of the reinforcement as defined in the ASME Code. If, because of wind loads or other factors, the longitudinal stresses govern, the least thickness in a similarly determined length of arc in the most critical plane perpendicular to the axis of the vessel also shall be averaged for computation of the longitudinal stresses. The thickness used for determining corrosion rates at the respective locations shall be the average thickness determined as in the preceding. For the purposes of 5.4, the actual thickness as determined by inspection shall be understood to mean the most critical value of the average thickness that has been determined.

e. Widely scattered pits may be ignored as long as the following are true:

1. No pit depth is greater than one-half the vessel wall thickness exclusive of the corrosion allowance.

2. The total area of the pits does not exceed 7 square inches (45 square centimeters) within any 8-inch (20-centimeter) diameter circle.

3. The sum of their dimensions along any straight line within the circle does not exceed 2 inches (5 centimeters).

f. As an alternative to the procedures just described, any components with thinning walls that, because of corrosion or other wastage, are below the minimum required wall thicknesses may be evaluated to determine if they are adequate for continued service. The thinning components may be evaluated by employing the design by analysis methods of Section VIII, Division 2, Appendix 4, of the ASME Code. These methods may also be used to evaluate blend ground areas where defects have been removed. It is important to ensure that there are no sharp corners in blend ground areas to minimize stress concentration effects.

When using this criteria, the stress value used in the original pressure vessel design shall be substituted for the S_m value of Division 2 if the design stress is less than or equal to ⅔-specified minimum yield strength (SMYS) at temperature. If the original design stress is greater than ⅔-specified minimum yield strength at temperature, then ⅔-specified minimum yield strength shall be substituted for S_m. When this approach is to be used, consulting with a pressure vessel engineer experienced in pressure vessel design is required.

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g. When the surface at a weld with a joint factor of other than 1.0, as well as surfaces remote from the weld, is corroded, an independent calculation using the appropriate weld joint factor must be made to determine if the thickness at the weld or remote from the weld governs the allowable working pressure. For this calculation, the surface at a weld includes 1 inch (2.5 centimeters) on either side of the weld or twice the minimum thickness on either side of the weld, whichever is greater.

h. When measuring the corroded thickness of ellipsoidal and torispherical heads, the governing thickness may be as follows:

1. The thickness of the knuckle region with the head rating calculated by the appropriate head formula.

2. The thickness of the central portion of the dished region, in which case the dished region may be considered a spherical segment whose allowable pressure is calculated by the code formula for spherical shells.

The spherical segment of both ellipsoidal and torispherical heads shall be considered to be that area located entirely within a circle whose center coincides with the center of the head and whose diameter is equal to 80 percent of the shell diameter. The radius of the dish of torispherical heads is to be used as the radius of the spherical segment (equal to the diameter of the shell for standard heads, though other radii have been permitted). The radius of the spherical segment of ellipsoidal heads shall be considered to be the equivalent spherical radius K_1D, where D is the shell diameter (equal to the major axis) and K_1 is given in Table 1. In Table 1, h is one-half the length of the minor axis [equal to the inside depth of the ellipsoidal head measured from the tangent line (headbend line)]. For many ellipsoidal heads, $D/2h$ equals 2.0.

Table 1—Values of Spherical Radius Factor K_1

$D/2h$	K_1
3.0	1.36
2.8	1.27
2.6	1.18
2.4	1.08
2.2	0.99
2.0	0.90
1.8	0.81
1.6	0.73
1.4	0.65
1.2	0.57
1.0	0.50

Note: The equivalent spherical radius equals K_1D; the axis ratio equals $D/2h$. Interpolation is permitted for intermediate values.

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6 Inspection and Testing of Pressure Vessels and Pressure-Relieving Devices

6.1 GENERAL

Pressure vessels shall be inspected at the time of installation. Internal field inspections of new vessels are not required as long as a manufacturer's data report assuring that the vessels are satisfactory for their intended service is available. To ensure vessel integrity, all pressure vessels shall be inspected at the frequencies provided in this section.

In selecting the technique(s) to be used for the inspection of a pressure vessel, both the condition of the vessel and the environment in which it operates should be taken into consideration. The inspection, as deemed necessary by the authorized pressure vessel inspector, may include many of a number of nondestructive techniques, including visual inspection. Internal inspection is preferred because process side degradation (corrosion, erosion, and environmental cracking) can be nonuniform throughout the vessel and, therefore, difficult to locate by external NDE. On-stream inspection may be acceptable in lieu of internal inspection for vessels under the specific circumstances defined in 6.4. In situations where on-stream inspection is acceptable, such inspection may be conducted either while the vessel is out of service and depressurized or on stream and under pressure. Except in response to an apparent need, such as when environmental cracking (see *Guide for Inspection of Refinery Equipment,* Chapter II) is suspected, inspection techniques exceeding the examination requirements used in the design and fabrication of the vessel are not required. 98

The appropriate inspection must provide the information necessary to determine that all of the essential sections or components of the vessel are safe to operate until the next scheduled inspection. The risks associated with operational shutdown and start-up and the possibility of increased corrosion due to exposure of vessel surfaces to air and moisture should be evaluated when an internal inspection is being planned.

6.2 RISK-BASED INSPECTION

Identifying and evaluating potential degradation mechanisms are important steps in an assessment of the likelihood of a pressure vessel failure. However, adjustments to inspection strategy and tactics to account for consequences of a failure should also be considered. Combining the assessment of likelihood of failure and the consequence of failure are essential elements of risk-based inspection (RBI). 98

When an owner/user chooses to conduct a RBI assessment, it must include a systematic evaluation of both the likelihood of failure and the associated consequence of failure. The likelihood assessment must be based on all forms of degradation that could reasonably be expected to affect a vessel in any particular service. Examples of those degradation mechanisms include: internal or external metal loss from an identified form of corrosion (localized or general), all forms of cracking, including hydrogen assisted and stress corrosion cracking (from the inside or outside surfaces of a vessel), and any other forms of metallurgical, corrosion, or mechanical degradation, such as fatigue, embrittlement, creep, etc. Additionally, the effectiveness of the inspection practices, tools, and techniques utilized for finding the expected and potential degradation mechanisms must be evaluated. This likelihood of failure assessment should be repeated each time equipment or process changes are made that could significantly affect degradation rates or cause premature failure of the vessel.

Other factors that should be considered in a RBI assessment include: appropriateness of the materials of construction; vessel design conditions, relative to operating conditions; appropriateness of the design codes and standards utilized; effectiveness of corrosion monitoring programs; and the quality of maintenance and inspection quality assurance/quality control programs. Equipment failure data and information will also be important information for this assessment. The consequence assessment must consider the potential incidents that may occur as a result of fluid release, including explosion, fire, toxic exposure, environmental impact, and other health effects associated with a failure of a vessel.

It is essential that all RBI assessments be thoroughly documented, clearly defining all the factors contributing to both the likelihood and consequence of a failure of the vessel. 98

After an effective RBI assessment is conducted, the results can be used to establish a vessel inspection strategy and more specifically better define the following:

a. The most appropriate inspection methods, scope, tools and techniques to be utilized based on the expected forms of degradation.
b. The appropriate frequency for internal, external, and on-stream inspections.
c. The need for pressure testing after damage has been incurred or after repairs or modifications have been completed.
d. The prevention and mitigation steps to reduce the likelihood and consequence of a vessel failure.

An RBI assessment may be used to increase or decrease the 10-year inspection limit described in Section 6.4. When used to increase the 10-year limit, RBI assessment shall be reviewed and approved by a pressure vessel engineer and authorized pressure vessel inspector at intervals not to exceed 10 years, or more often if warranted by process, equipment, or consequence changes.

6.3 EXTERNAL INSPECTION 98

Each vessel aboveground shall be given a visual external inspection, preferably while in operation, at least every 5 years or at the same interval as the required internal or on-stream inspection, whichever is less. The inspection shall, at the least,

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determine the condition of the exterior insulation, the condition of the supports, the allowance for expansion, and the general alignment of the vessel on its supports. Any signs of leakage should be investigated so that the sources can be established. Inspection for corrosion under insulation (CUI) shall be considered for externally-insulated vessels subject to moisture ingress and that operate between 25°F (–4°C) and 250°F (120°C), or are in intermittent service. This inspection may require removal of some insulation. It is not normally necessary to remove insulation if the entire vessel shell is always operated at a temperature sufficiently low [below 25°F (–4°C)] or sufficiently high [above 250°F (120°C)] to prevent the presence or condensation of moisture under the insulation. Alternatively, shell thickness measurements done internally at typical problem areas (for example, stiffening rings, around nozzles, and other locations which tend to trap moisture or allow moisture ingress) may be performed during internal inspections.

Buried vessels shall be inspected to determine their external environmental condition. The inspection interval shall be based on corrosion-rate information obtained from one or more of the following methods: (a) during maintenance activity on adjacent connecting piping of similar material; (b) from the interval examination (specified in the paragraph above) of similarly buried corrosion test coupons of similar material; (c) from representative portions of the actual vessel; or (d) from a vessel in similar circumstances.

Vessels that are known to have a remaining life of over 10 years or that are protected against external corrosion—for example, (a) vessels insulated effectively to preclude the entrance of moisture, (b) jacketed cryogenic vessels, (c) vessels installed in a cold box in which the atmosphere is purged with an inert gas, and (d) vessels in which the temperature being maintained is sufficiently low or sufficiently high to preclude the presence of water—do not need to have insulation removed for the external inspection. However, the condition of their insulating system or their outer jacketing, such as the cold box shell, shall be observed at least every 5 years and repaired if necessary.

6.4 INTERNAL AND ON-STREAM INSPECTION

The period between internal or on-stream inspections shall not exceed one half the estimated remaining life of the vessel based on corrosion rate or 10 years, whichever is less. In cases where the remaining safe operating life is estimated to be less than 4 years, the inspection interval may be the full remaining safe operating life up to a maximum of 2 years.

For pressure vessels that are in noncontinuous service and are isolated from the process fluids such that they are not exposed to corrosive environments (such as inert gas purged or filled with noncorrosive hydrocarbons), the 10 years shall be the 10 years of actual service exposed life. Equipment that is not adequately protected from corrosive environments may experience significant internal corrosion while idle and should be carefully reviewed when setting inspection intervals. In no case should these exceed one-half the estimated remaining corrosion-rate life, or 10 years since the last inspection. External inspections for vessels in noncontinuous service remain the same as for continuous service, as outlined in 6.3.

Except as noted below, internal inspection is normally the preferred method of inspection and shall be conducted on vessels subject to significant localized corrosion and other types of damage. At the discretion of the authorized pressure vessel inspector, on-stream inspection may be substituted for internal inspection in the following situations:

a. When size, configuration, or lack of access makes vessel entry for internal inspection physically impossible.

b. When the general corrosion rate of a vessel is known to be less than 0.005 inch (0.125 millimeter) per year and the estimated remaining life is greater than 10 years, and all of the following conditions are met:

1. The corrosive character of the contents, including the effect of trace components, has been established by at least 5 years of the same or comparable service experience with the type of contents being handled.
2. No questionable condition is disclosed by the external inspection specified in 6.3.
3. The operating temperature of the steel vessel shell does not exceed the lower temperature limits for the creep-rupture range of the vessel material.
4. The vessel is not considered to be subject to environmental cracking or hydrogen damage from the fluid being handled.
5. The vessel is not strip-lined or plate-lined.

If the requirements of item b above are not met, as a result of conditions noted during the scheduled on-stream inspection, the next scheduled inspection shall be an internal inspection. When a vessel has been internally inspected, the results of this inspection can be used to determine whether an on-stream inspection can be substituted for an internal inspection on a similar vessel operating in the same service and conditions.

When an on-stream inspection is conducted in lieu of an internal inspection, a thorough examination shall be performed using ultrasonic thickness measurements, or radiography, or other appropriate means of NDE to measure metal thicknesses and/or assess the integrity of the metal and welds. If an on-stream inspection is conducted, the authorized pressure vessel inspector shall be given sufficient access to all parts of the vessel (heads, shell, and nozzles) so that the inspector is satisfied that an accurate assessment of the vessel condition can be made.

A representative number of thickness measurements must be conducted on each vessel to satisfy the requirements for an internal or on-stream inspection. For example, the thickness for all major components (shells, heads, cone sections) and a rep-

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resentative sample of vessel nozzles should be measured and recorded, and the remaining life and next inspection interval should be calculated for the limiting component. A decision on the number and location of the thickness measurements should consider results from previous inspections, if available, and the potential consequence of loss of containment. Measurements at a number of thickness measurement locations (TMLs) are intended to establish general and localized corrosion rates in different sections of the vessel. A minimal number of TMLs are acceptable when the established rate of corrosion is low and not localized. For pressure vessels susceptible to localized corrosion, it is vital that those knowledgeable in localized corrosion mechanisms be consulted about the appropriate placement and number of TMLs. Additionally, for localized corrosion, it is important that inspections are conducted using scanning methods such as profile radiography, scanning ultrasonics, and/or other suitable NDE methods that will reveal the scope and extent of localized corrosion.

The remaining life of the vessel shall be calculated from the following formula:

$$\text{Remaining life (years)} = \frac{t_{\text{actual}} - t_{\text{minimum}}}{\text{corrosion rate}}$$

[inches (millimeters) per year]

Where:

t_{actual} = the thickness, in inches (millimeters), recorded at the time of inspection for a given location or component.

t_{minimum} = minimum allowable thickness, in inches (millimeters), for a given location or component.

$$\text{Corrosion rate} = \frac{t_{\text{previous}} - t_{\text{actual}}}{\text{years between } t_{\text{previous}} \text{ and } t_{\text{actual}}}$$

t_{previous} = the thickness, in inches (millimeters), at the same location as t_{actual} measured during a previous inspection.

A statistical analysis may be used in the corrosion rate and remaining life calculations for the pressure vessel sections. This statistical approach may be applied for assessment of substituting an internal inspection (item b in the preceding), or for determining the internal inspection interval. Care must be taken to ensure that the statistical treatment of data results reflects the actual condition of the vessel section. Statistical analysis is not applicable to vessels with significant localized corrosion.

The determination of corrosion rate may include thickness data collected at more than two different times. Suitable use of short-term versus long-term corrosion rates shall be determined by the authorized pressure vessel inspector. When there is a discrepancy between short-term and long-term corrosion rates, a pressure vessel engineer experienced in corrosion may need to be consulted about the use of these rates, at the discretion of the inspector, for calculating the remaining life and next inspection date.

For a large vessel with two or more zones of differing corrosion rates, each zone may be treated independently regarding the interval between inspections or for substituting the internal inspection with an on-stream inspection. If a multi-zone analysis is used, the zone with the shortest remaining half-life shall be used as the limiting case for setting the internal inspection interval or for substituting the internal inspection with an on-stream inspection.

An alternative method to establish the required inspection interval based on remaining life is by calculation of the projected maximum allowable working pressure (MAWP) of each vessel component as described in 5.4. This procedure may be iterative involving selection of an inspection interval, determination of the corrosion loss expected over the interval, and calculation of the projected MAWP. The inspection interval is within the maximum permitted as long as the projected MAWP of the limiting component is not less than the lower of the name plate or rerated MAWP. The maximum inspection interval using this method is also 10 years.

When problems are experienced with external loading, faulty material, or fabrication the remaining life as determined above shall be reduced to recognize those conditions. If deterioration due to conditions such as those mentioned in 5.2 is detected, the inspection interval must be appropriately adjusted.

If the service conditions of a vessel are changed, the maximum operating pressure, the maximum and minimum operating temperature, and the period of operation until the next inspection shall be established for the new service conditions.

If both the ownership and the location of a vessel are changed, the vessel shall be internally and externally inspected before it is reused, and the allowable conditions of service and the next period of inspection shall be established for the new service.

6.5 PRESSURE TEST

98

When the authorized pressure vessel inspector believes that a pressure test is necessary or when, after certain repairs or alterations, the inspector believes that one is necessary (see 7.2.9), the test shall be conducted at a pressure in accordance with the construction code used for determining the maximum allowable working pressure. To minimize the risk of brittle fracture during the test, the metal temperature should be maintained at least 30°F (17°C) above the minimum design metal temperature for vessels that are more than 2 inches (5 centimeters) thick, or 10°F (6°C) above for vessels that have a thickness of 2 inches (5 centimeters) or less. The test temperature need not exceed 120°F (50°C) unless there is information on the brittle characteristics of the vessel material indicating that a lower test temperature is acceptable or a higher test temperature is needed.[5]

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Pneumatic testing may be used when hydrostatic testing is impracticable because of temperature, foundation, refractory lining, or process reasons; however, the potential personnel and property risks of pneumatic testing shall be considered before such testing is carried out. As a minimum, the inspection precautions contained in the ASME Code shall be applied in any pneumatic testing. Before applying a hydrostatic test to equipment, consideration should be given to the supporting structure and the foundation design.

When a pressure test is to be conducted in which the test pressure will exceed the set pressure of the safety relief valve with the lowest setting, the safety relief valve or valves should be removed. An alternative to removing the safety relief valves is to use test clamps to hold down the valve disks. Applying an additional load to the valve spring by turning the compression screw is not recommended. Other appurtenances, such as gauge glasses, pressure gauges, and rupture disks, that may be incapable of withstanding the test pressure should also be removed or should be blanked off or vented. When the pressure test has been completed, pressure relief devices of the proper settings and other appurtenances removed or made inoperable during the pressure test shall be reinstalled or reactivated.

98 | 6.6 PRESSURE-RELIEVING DEVICES

Pressure relief valves shall be tested and repaired by repair organizations experienced in valve maintenance. Each repair organization shall have a fully documented quality control system. As a minimum, the following requirements and pieces of documentation should be included in the quality control system:

98 a. Title page.
b. Revision log.
c. Contents page.
d. Statement of authority and responsibility.
e. Organizational chart.
f. Scope of work.
g. Drawings and specification controls.
h. Material and part control.
i. Repair and inspection program.
j. Welding, nondestructive examination, and heat treatment procedures.
k. Valve testing, setting, leak testing, and sealing.
l. General example of the valve repair nameplate.
m. Procedures for calibrating measurement and test gauges.
n. Controlled copies of the manual.
o. Sample forms.
p. Repair personnel training or qualifications.

[5]For vessels without minimum design metal temperature, the minimum acceptable operating temperature should be used in lieu of the minimum design metal temperature.

Each repair organization shall also have a fully documented training program that shall ensure that repair personnel are qualified within the scope of the repairs.

Pressure relief valves shall be tested at intervals that are frequent enough to verify that the valves perform reliably. This may include testing pressure relief valves on newly installed equipment. Pressure-relieving devices should be tested and maintained in accordance with API Recommended Practice 576. Other pressure-relieving devices, such as rupture disks and vacuum-breaker valves, shall be thoroughly examined at intervals determined on the basis of service. 98

The intervals between pressure-relieving-device testing or inspection should be determined by the performance of the devices in the particular service concerned. Test or inspection intervals on pressure-relieving devices in typical process services should not exceed 5 years, unless service experience indicates that a longer interval is acceptable. For clean (nonfouling), noncorrosive services, maximum intervals may be increased to 10 years. When service records indicate that a pressure-relieving device was heavily fouled or stuck in the last inspection or test, the service interval shall be reduced if the review shows that the device may not perform reliably in the future. The review should include an effort to determine the cause of the fouling or the reasons for the relief device not operating properly.

6.7 RECORDS 98

Pressure vessel owners and users shall maintain permanent and progressive records of their pressure vessels. Permanent records will be maintained throughout the service life of each vessel; progressive records will be regularly updated to include new information pertinent to the operation, inspection, and maintenance history of the vessel.

Pressure vessel records shall contain three types of vessel information pertinent to mechanical integrity as follows:

a. Construction and design information. For example, equipment serial number or other identifier, manufacturers' data reports (MDRs), design specification data, design calculations (where MDRs are unavailable), and construction drawings.

b. Operating and inspection history. For example, operating conditions, including process upsets that may affect mechanical integrity, inspection reports, and data for each type of inspection conducted (for example, internal, external, thickness measurements), and inspection recommendations for repair. See Appendix C for sample pressure vessel inspection records. Inspection reports shall document the date of each inspection and/or test, the date of the next scheduled inspection, the name of the person who performed the inspection and/or test, the serial number or other identifier of the equipment inspected, a description of the inspec-

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tion and/or test performed, and the results of the inspection and/or test.

c. Repair, alteration, and rerating information. For example, (1) repair and alteration forms like that shown in Appendix D, (2) reports indicating that equipment still in-service with identified deficiencies or recommendations for repair are suitable for continued service until repairs can be completed, and (3) rerating documentation (including rerating calculations, new design conditions, and evidence of stamping).

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2. External inspections shall be performed when an on-stream or internal inspection is performed or at shorter intervals at the owner or user's option.

3. On-stream or internal inspections shall be performed at least every 15 years or ¾-remaining corrosion-rate life, whichever is less.

4. Any signs of leakage or deterioration detected in the interval between inspections shall require an on-stream or internal inspection of that vessel and a reevaluation of the inspection interval for that vessel class.

c. Higher-risk vessels shall be inspected as follows:

1. External inpections shall be performed when an on-stream or internal inspection is performed or at shorter intervals at the owner or user's option.

2. On-stream or internal inspections shall be performed at least every 10 years or ½-remaining corrosion rate life, which ever is less.

3. In cases where the remaining life is estimated to be less than 4 years, the inspection interval may be the full remaining life up to a maximum of 2 years. Consideration should also be given to increasing the number of vessels inspected within that class to improve the likelihood of detecting the worst-case corrosion.

4. Any signs of leakage or deterioration detected in the interval between inspections shall require an on-stream or internal inspection of that vessel and a reevaluation of the inspection interval for that vessel class.

d. Pressure vessels (whether grouped into classes or not) shall be inspected at intervals sufficient to insure their fitness for continued service. Operational conditions and vessel integrity may require inspections at shorter intervals than the intervals stated above.

e. If service conditions change, the maximum operating temperature, pressure, and interval between inspections must be reevaluated.

f. For large vessels with two or more zones of differing corrosion rates, each zone may be treated independently regarding the interval between inspections.

8.3.6 Additional Inspection Requirements

Additional inspection requirements, regardless of vessel classification, exist for the following vessels:

a. Vessels that have changed ownership and location must have an on-stream or internal inspection performed to establish the next inspection interval and to assure that the vessel is suitable for its intended service. Inspection of new vessels is not required if a manufacturer's data report is available.

b. If a vessel is transferred to a new location, and it has been more than 5 years since the vessel's last inspection, an on-stream or internal inspection is required. (Vessels in truck-mounted, skid-mounted, ship-mounted, or barge-mounted equipment are not included.)

c. Air receivers (other than portable equipment) shall be inspected at least every 5 years.

d. Portable or temporary pressure vessels that are employed for the purpose of testing oil and gas wells during completion or recompletion shall be inspected at least once during each 3-year period of use. More frequent inspections shall be conducted if vessels have been in severe corrosive environments.

8.4 PRESSURE TEST

When a pressure test is conducted, the test shall be in accordance with the procedures in 6.5. |98

8.5 SAFETY RELIEF DEVICES

Safety relief devices shall be inspected, tested, and repaired in accordance with 6.6. |98

8.6 RECORDS

The following records requirements apply:

a. Pressure vessel owners and users shall maintain pressure vessel records. The preferred method of record keeping is to maintain data by individual vessel. Where vessels are grouped into classes, data may be maintained by vessel class. When inspections, repairs, or alterations are made on an individual vessel, specific data shall be recorded for that vessel.

b. Examples of information that may be maintained are vessel identification numbers; safety relief device information; and the forms on which results of inspections, repairs, alterations, or reratings are to be recorded. Any appropriate forms may be used to record these results. A sample pressure vessel inspection record is shown in Appendix C. A sample alteration or rerating of pressure vessel form is shown in Appendix D. Information on maintenance activities and events that affect vessel integrity should be included in the vessel records.

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APPENDIX B—AUTHORIZED PRESSURE VESSEL INSPECTOR CERTIFICATION

B.1 Examination

A written examination to certify inspectors within the scope of API 510, *Pressure Vessel Inspection Code: Maintenance Inspection, Rating, Repair, and Alteration,* shall be administered by API or a third party designated by API. The examination shall be based on the current *API 510 Inspector Certification Body of Knowledge* as published by API. 98

B.2 Certification

B.2.1 An API 510 authorized pressure vessel inspector certificate will be issued when an applicant has successfully passed the API 510 certification examination and satisfies the criteria for education and experience. His/her education and experience, when combined, shall be equal to at least one of the following:

a. A Bachelor of Science degree in engineering or technology, plus one year of experience in supervision of inspection activities or performance of inspection activities as described in API 510.

b. A two-year degree or certificate in engineering or technology, plus two years of experience in the design, construction, repair, inspection, or operation of pressure vessels, of which one year must be in supervision of inspection activities or performance of inspection activities as described in API 510. 98

c. A high school diploma or equivalent, plus three years of experience in the design, construction, repair, inspection, or operation of pressure vessels, of which one year must be in supervision of inspection activities or performance of inspection activities as described in API 510.

d. A minimum of five years of experience in the design, construction, repair, inspection, or operation of pressure vessels, of which one year must be in supervision of inspection activities or performance of inspection activities as describe in API 510.

B.2.2 An API 510 authorized pressure vessel inspector certificate may be issued when an applicant provides documented evidence of passing the National Board of Boiler and Pressure Vessel Inspectors examination and meets all requirements for education and experience of API 510.

B.2.3 An API certificate for an authorized pressure vessel inspector is valid for three years from its date of issuance.

B.2.4 An API 510 authorized pressure vessel inspector certification is valid in all jurisdictions and any other location that accepts or otherwise does not prohibit the use of API 510. 98

B.3 Certification Agency

The American Petroleum Institute shall be the certifying agency. 98

B.4 Retroactivity

The certification requirements of API 510 shall not be retroactive or interpreted as applying before 12 months after the date of publication of this edition or addendum of API 510. The recertification requirements of API 510 paragraph B.5.2 shall not be retroactive or interpreted as applying before 3 years after the date of publication of this edition or addendum of API 510. 98

B.5 Recertification

B.5.1 Recertification is required 3 years from the date of issuance of the API 510 authorized pressure vessel inspector certificate. Recertification by written test will be required for authorized pressure vessel inspectors who have not been actively engaged as authorized pressure vessel inspectors within the previous 3 years. Exams will be in accordance with all provisions contained in API 510.

B.5.2 "Actively engaged as an authorized pressure vessel inspector" shall be defined by one of the following provisions: 98

a. A minimum of 20% of time spent performing inspection activities or supervision inspection activities as described in the API 510 inspection code over the most recent 3-year certification period.

b. Performance of inspection activities or supervision of inspection activities on 75 pressure vessels as described in API 510 over the most recent 3-year certification period.

Note: Inspection activities common to other API inspection documents (NDE, record-keeping, review of welding documents, etc.) may be considered here.

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Pressure Vessel Inspection Code: Maintenance Inspection, Rating, Repair, and Alteration

API 510
EIGHTH EDITION, JUNE 1997



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Pressure Vessel Inspection Code: Maintenance Inspection, Rating, Repair, and Alteration

Manufacturing, Distribution and Marketing Department

API 510
EIGHTH EDITION, JUNE 1997



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FOREWORD

In December 1931, API and the American Society of Mechanical Engineers (ASME) created the Joint API/ASME Committee on Unfired Pressure Vessels. This committee was created to formulate and prepare for publication a code for safe practices in the design, construction, inspection, and repair of pressure vessels to be used in the petroleum industry. Entitled API/ASME *Code for Unfired Pressure Vessels for Petroleum Liquids and Gases* (commonly called the API/ASME *Code for Unfired Pressure Vessels* or API/ASME Code), the first edition of the code was approved for publication in 1934.

From its inception, the API/ASME Code contained Section I, which covered recommended practices for vessel inspection and repair and for establishing allowable working pressures for vessels in service. Section I recognized and afforded well-founded bases for handling various problems associated with the inspection and rating of vessels subject to corrosion. Although the provisions of Section I (like other parts of the API/ASME Code) were originally intended for pressure vessels installed in the plants of the petroleum industry, especially those vessels containing petroleum gases and liquids, these provisions were actually considered to be applicable to pressure vessels in most services. ASME's Boiler and Pressure Vessel Committee adopted substantially identical provisions and published them as a nonmandatory appendix in the 1950, 1952, 1956, and 1959 editions of Section VIII of the ASME *Boiler and Pressure Vessel Code.*

After the API/ASME Code was discontinued in 1956, a demand arose for the issuance of Section I as a separate publication, applicable not only to vessels built in accordance with any edition of the API/ASME Code but also to vessels built in accordance with any edition of Section VIII of the ASME Code. Such a publication appeared to be necessary to assure industry that the trend toward uniform maintenance and inspection practices afforded by Section I of the API/ASME Code would be preserved. API 510, first published in 1958, is intended to satisfy this need.

The procedures in Section I of the 1951 edition of the API/ASME Code, as amended by the March 16, 1954 addenda, have been updated and revised in API 510. Section I of the API/ASME Code contained references to certain design or construction provisions, so these references have been changed to refer to provisions in the ASME Code. Since the release of the 1960 edition of the *National Board Inspection Code*, elements of the API/ASME Code have also been carried by the *National Board Inspection Code.*

It is the intent of API to keep this publication up to date. All pressure vessel owners and operators are invited to report their experiences in the inspection and repair of pressure vessels whenever such experiences may suggest a need for revising or expanding the practices set forth in API 510.

This edition of API 510 supersedes all previous editions of API 510, *Pressure Vessel Inspection Code: Maintenance Inspection, Rating, and Repair of Pressure Vessels.*

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IMPORTANT INFORMATION CONCERNING USE OF ASBESTOS OR ALTERNATIVE MATERIALS

Asbestos is specified or referenced for certain components of the equipment described in some API standards. It has been of extreme usefulness in minimizing fire hazards associated with petroleum processing. It has also been a universal sealing material, compatible with most refining fluid services.

Certain serious adverse health effects are associated with asbestos, among them the serious and often fatal diseases of lung cancer, asbestosis, and mesothelioma (a cancer of the chest and abdominal linings). The degree of exposure to asbestos varies with the product and the work practices involved.

Consult the most recent edition of the Occupational Safety and Health Administration (OSHA), U.S. Department of Labor, Occupational Safety and Health Standard for Asbestos, Tremolite, Anthophyllite, and Actinolite, 29 *Code of Federal Regulations* Section 1910.1001; the U.S. Environmental Protection Agency, National Emission Standard for Asbestos, 40 *Code of Federal Regulations* Sections 61.140 through 61.156; and the U.S. Environmental Protection Agency (EPA) rule on labeling requirements and phased banning of asbestos products, published at 54 *Federal Register* 29460 (July 12, 1989).

There are currently in use and under development a number of substitute materials to replace asbestos in certain applications. Manufacturers and users are encouraged to develop and use effective substitute materials that can meet the specifications for, and operating requirements of, the equipment to which they would apply.

SAFETY AND HEALTH INFORMATION WITH RESPECT TO PARTICULAR PRODUCTS OR MATERIALS CAN BE OBTAINED FROM THE EMPLOYER, THE MANUFACTURER OR SUPPLIER OF THAT PRODUCT OR MATERIAL, OR THE MATERIAL SAFETY DATA SHEET.

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Pressure Vessel Inspection Code: Maintenance Inspection, Rating, Repair, and Alteration

1 Scope

1.1 GENERAL APPLICATION

This inspection code covers the maintenance inspection, repair, alteration, and rerating procedures for pressure vessels used by the petroleum and chemical process industries. The application of this inspection code is restricted to organizations that employ or have access to an authorized inspection agency as defined in 3.4. Except as provided in 1.2, the use of this inspection code is restricted to organizations that employ or have access to engineering and inspection personnel or organizations that are technically qualified to maintain, inspect, repair, alter, or rerate pressure vessels. Pressure vessel inspectors are to be certified as stated in this inspection code. Since other codes that cover specific industries and general service applications already exist (for example, Sections VI, VII, and XI of the ASME *Boiler and Pressure Vessel Code* and the *National Board Inspection Code*), the industries that fit within the restrictions above have developed this inspection code to fulfill their own specific requirements.

This inspection code applies to vessels constructed in accordance with the API/ASME *Code for Unfired Pressure Vessels for Petroleum Liquids and Gases*, Section VIII of the ASME Code, and other recognized pressure vessel codes; to nonstandard vessels; and to other vessels constructed noncode or approved as jurisdictional special. This inspection code is only applicable to vessels that have been placed in service (including items further described in 1.2) and have been inspected by an authorized inspection agency or repaired by a repair organization as defined in 3.15.

Adoption and use of this inspection code does not permit its use in conflict with any prevailing regulatory requirements.

1.2 SPECIFIC APPLICATIONS

1.2.1 All pressure vessels used for Exploration and Production (E&P) service [for example, drilling, producing, gathering, transporting, lease processing, and treating liquid petroleum, natural gas, and associated salt water (brine)] may be inspected under the alternative rules set forth in Section 8. Except for Section 6, all of the sections in this inspection code are applicable to pressure vessels in E&P service. The alternative rules in Section 8 are intended for services that may be regulated under safety, spill, emission, or transportation controls by the U.S. Coast Guard; the Office of Hazardous Materials Transportation of the U.S. Department of Transportation (DOT) and other units of DOT; the Minerals Management Service of the U.S. Department of the Interior; state and local oil and gas agencies; or any other regulatory commission.

1.2.2 The following are excluded from the specific requirements of this inspection code:

a. Pressure vessels on movable structures covered by other jurisdictional regulations (see Appendix A).
b. All classes of containers listed for exemption from construction in the scope of Section VIII, Division 1, of the ASME Code (see Appendix A).
c. Pressure vessels that do not exceed the following volumes and pressures:
   1. Five cubic feet (0.141 cubic meters) in volume and 250 pounds per square inch (1723.1 kilopascals) design pressure.
   2. One and a half cubic feet (0.042 cubic meters) in volume and 600 pounds per square inch (4136.9 kilopascals) design pressure (see Appendix A).

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2 References

The most recent editions of the following standards, codes, and specifications are cited in this inspection code.

API

RP 572 *Inspection of Pressure Vessels*

RP 574 *Inspection of Piping, Tubing, Valves, and Fittings*

RP 576 *Inspection of Pressure-Relieving Devices*

Guide for Inspection of Refinery Equipment, Chapter II, "Conditions Causing Deterioration or Failures"

Note: This publication is out of print. To obtain a copy please inform the person taking your order that you require this publication for the API 510 Inspector Certification Exam.

ASME[1]

Boiler and Pressure Vessel Code, Section V, Section VI, Section VII, Section VIII, Section IX, and Section XI

National Board[2]

National Board Inspection Code

[1]American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.

[2]National Board of Boiler and Pressure Vessel Inspectors, 1055 Crupper Avenue, Columbus, Ohio 43229.

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3 Definitions

For the purposes of this standard, the following definitions apply.

3.1 alteration: A physical change in any component or a rerating that has design implications that affect the pressure-containing capability of a pressure vessel beyond the scope of the items described in existing data reports. The following should not be considered alterations: any comparable or duplicate replacement, the addition of any reinforced nozzle less than or equal to the size of existing reinforced nozzles, and the addition of nozzles not requiring reinforcement.

3.2 ASME Code: Abbreviation and shortened title for the ASME *Boiler and Pressure Vessel Code*. This abbreviated title includes the addenda and code cases of the ASME *Boiler and Pressure Vessel Code*.

The ASME Code is written for new construction; however, most of the technical requirements for design, welding, examination, and materials can be applied in the maintenance inspection, rating, repair, and alteration of operating pressure vessels. When the ASME Code cannot be followed because of its new construction orientation (new or revised material specifications, inspection requirements, certain heat treatments and pressure tests, and stamping and inspection requirements), the engineer or inspector shall conform to this inspection code rather than to the ASME Code. If an item is covered by requirements in the ASME Code and this inspection code or if there is a conflict between the two codes, for vessels that have been placed in service, the requirements of this inspection code shall take precedence over the ASME Code. As an example of the intent of this inspection code, the phrase "applicable requirements of the ASME Code" has been used in this inspection code instead of the phrase "in accordance with the ASME Code."

3.3 authorized pressure vessel inspector: An employee of an authorized inspection agency who is qualified and certified to perform inspections under this inspection code.

3.4 authorized inspection agency: Any one of the following:

a. The inspection organization of the jurisdiction in which the pressure vessel is used.
b. The inspection organization of an insurance company that is licensed or registered to write and actually does write pressure vessel insurance.
c. The inspection organization of an owner or user of pressure vessels who maintains an inspection organization for his equipment only and not for vessels intended for sale or resale.
d. An independent organization or individual that is under contract to and under the direction of an owner-user and that is recognized or otherwise not prohibited by the jurisdiction in which the pressure vessel is used. The owner-user's inspection program shall provide the controls that are necessary when contract inspectors are used.

3.5 construction code: The code or standard to which a vessel was originally built, such as API/ASME, API, or State Special/non-ASME.

3.6 inspection code: Shortened title for API 510 used in this publication.

3.7 jurisdiction: A legally constituted government administration that may adopt rules relating to pressure vessels.

3.8 maximum allowable working pressure: The maximum gauge pressure permitted at the top of a pressure vessel in its operating position for a designated temperature. This pressure is based on calculations using the minimum (or average pitted) thickness for all critical vessel elements, exclusive of thickness designated for corrosion and loadings other than pressure.

3.9 minimum allowable shell thickness: The thickness required for each element of a vessel. The minimum allowable shell thickness is based on calculations that consider temperature, pressure, and all loadings.

3.10 on-stream inspection: The inspection used to establish the suitability of a pressure vessel for continued operation. Nondestructive examination (NDE) procedures are used to establish the suitability of the vessel, and the vessel may or may not be in operation while the inspection is being carried out. Because a vessel may be in operation while an on-stream inspection is being carried out, an on-stream inspection means essentially that the vessel is not entered for internal inspection.

3.11 pressure vessel: A container designed to withstand internal or external pressure. This pressure may be imposed by an external source, by the application of heat from a direct or indirect source, or by any combination thereof. This definition includes unfired steam generators and other vapor generating vessels which use heat from the operation of a processing system or other indirect heat source. (Specific limits and exemptions of equipment covered by this inspection code are given in Section 1 and Appendix A.)

3.12 pressure vessel engineer: Shall be one or more persons or organizations acceptable to the owner-user who are knowledgeable and experienced in the engineering disciplines associated with evaluating mechanical and material characteristics which affect the integrity and reliability of pressure vessels. The pressure vessel engineer, by consulting with appropriate specialists, should be regarded as a composite of all entities needed to properly assess the technical requirements.

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3.13 quality assurance: All planned, systematic, and preventative actions required to determine if materials, equipment, or services will meet specified requirements so that equipment will perform satisfactorily in service. The contents of a quality assurance inspection manual are outlined in 4.3.

3.14 repair: The work necessary to restore a vessel to a condition suitable for safe operation at the design conditions. If any repair changes the design temperature or pressure, the requirements for rerating shall be satisfied. A repair can be the addition or replacement of pressure or nonpressure parts that do not change the rating of the vessel.

3.15 repair organization: Any one of the following:

a. The holder of a valid ASME Certificate of Authorization that authorizes the use of an appropriate ASME Code symbol stamp.
b. An owner or user of pressure vessels who repairs his or her own equipment in accordance with this inspection code.
c. A contractor whose qualifications are acceptable to the pressure-vessel owner or user and who makes repairs in accordance with this inspection code.
d. An individual or organization that is authorized by the legal jurisdiction.

3.16 rerating: A change in either the temperature ratings or the maximum allowable working pressure rating of a vessel, or a change in both. The maximum allowable working temperature and pressure of a vessel may be increased or decreased because of a rerating, and sometimes a rerating requires a combination of changes. Derating below original design conditions is a permissible way to provide for corrosion. When a rerating is conducted in which the maximum allowable working pressure or temperature is increased or the minimum temperature is decreased so that additional mechanical tests are required, it shall be considered an alteration.

3.17 examiner: A person who assists the API authorized pressure vessel inspector by performing specific NDE on pressure vessels but does not evaluate the results of those examinations in accordance with API 510, unless specifically trained and authorized to do so by the owner or user. The examiner need not be certified in accordance with API 510 or be an employee of the owner or user but shall be trained and competent in the applicable procedures in which the examiner is involved. In some cases, the examiner may be required to hold other certifications as necessary to satisfy the owner or user requirements. Examples of other certification that may be required are American Society for Nondestructive Testing[3] SNT-TC-1A, or CP189, or American Welding Society[4] Welding Inspector Certification. The examiner's employer shall maintain certification records of the examiners employed, including dates and results of personnel qualifications and shall make them available to the API authorized pressure vessel inspector.

[3]American Society for Nondestructive Testing, Inc., 1711 Arlingate Lane, P.O. Box 28518, Columbus, Ohio, 43228-0518.
[4]American Welding Society, 550 N.W. LeJeune Road, Miami, FL 33135.

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4 Owner-User Inspection Organization

4.1 GENERAL

An owner or user of pressure vessels who controls the frequency of the inspections of the pressure vessels or the maintenance of them is responsible for the functions of an authorized inspection agency, as stated in the provisions of this inspection code. This owner-user inspection organization may also control activities relating to the maintenance inspection, rating, repair, and alteration of these pressure vessels.

4.2 API AUTHORIZED PRESSURE VESSEL INSPECTOR QUALIFICATION AND CERTIFICATION

An authorized pressure vessel inspector employed by or under contract to and under the direction of an owner-user inspection organization shall be educated and experienced. His education and experience, when combined, shall be equal to at least one of the following:

a. A degree in engineering plus 1 year of experience in the design, construction, repair, operation, or inspection of boilers or pressure vessels.
b. A 2-year certificate in engineering or technology from a technical college plus 2 years of experience in the design, construction, repair, operation, or inspection of boilers or pressure vessels.
c. The equivalent of a high school education plus 3 years of experience in the construction, repair, operation, or inspection of boilers or pressure vessels.
d. Five years of experience in the inspection of boilers or pressure vessels.

In addition, the authorized pressure vessel inspector shall be certified by an agency as provided in this inspection code (see Appendix B).

4.3 OWNER-USER ORGANIZATION RESPONSIBILITIES

An owner-user organization is responsible for developing, documenting, implementing, executing, and assessing pressure vessel inspection systems and inspection procedures that will meet the requirements of this inspection code. These systems and procedures will be contained in a quality assurance inspection manual and shall include the following:

a. Organization and reports of structure for inspection personnel.
b. Documentation and maintenance of inspection and quality assurance procedures.
c. Documentation and reports of inspection and test results.
d. Corrective action for inspection and test results.
e. Internal audits for compliance with the quality assurance inspection manual.
f. Review and approval of drawings, design calculations, and specifications for repairs, alterations, and reratings.
g. Assurance that all jurisdictional requirements for pressure vessel inspection, repairs, alterations, and rerating are continuously met.
h. Reports to the authorized pressure vessel inspector any process changes that could affect pressure vessel integrity.
i. Training requirements for inspection personnel regarding inspection tools, techniques, and technical knowledge base.
j. Controls necessary so that only qualified welders and procedures are used for all repairs and alterations.
k. Controls necessary so that only qualified nondestructive examination (NDE) personnel and procedures are utilized.
l. Controls necessary so that only materials conforming to the applicable section of the ASME Code are utilized for repairs and alterations.
m. Controls necessary so that all inspection measurement and test equipment are properly maintained and calibrated.
n. Controls necessary so that the work of contract inspection or repair organizations meet the same inspection requirements as the owner-user organization.
o. Internal auditing requirements for the quality control system for pressure-relieving devices.

4.4 API AUTHORIZED PRESSURE VESSEL INSPECTOR DUTY

When inspections, repairs, or alterations are being conducted on pressure vessels, an API authorized pressure vessel inspector shall be responsible to the owner-user for determining that the requirements of API 510 on inspection, examination, and testing are met, and shall be directly involved in the inspection activities. The API authorized pressure vessel inspector may be assisted in performing-visual inspections by other properly trained and qualified individuals, who may or may not be certified vessel inspectors. Personnel performing nondestructive examinations shall meet the qualifications identified in 3.17 but need not be API authorized pressure vessel inspectors. However, all examination results must be evaluated and accepted by the API authorized pressure vessel inspector.

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5 Inspection Practices

5.1 PREPARATORY WORK

Safety precautions are important in pressure vessel inspection because of the limited access to and the confined spaces of pressure vessels. Occupational Safety and Health Administration (OSHA) regulations pertaining to confined spaces and any other OSHA safety rules should be reviewed and followed, where applicable.

For an internal inspection, the vessel should be isolated by blinds or other positive methods from all sources of liquids, gases, or vapors. The vessel should be drained, purged, cleaned, ventilated, and gas tested before it is entered. Where required, protective equipment should be worn that will protect the eyes, lungs, and other parts of the body from specific hazards that may exist in the vessel.

The nondestructive testing equipment used for the inspection is subject to the safety requirements customarily followed in a gaseous atmosphere. Before the inspection is started, all persons working around the vessel should be informed that people are going to be working inside it. People working inside the vessel should be informed when any work is going to be done on the exterior of it.

The tools and personnel safety equipment needed for the vessel inspection should be checked before the inspection. Other equipment that might be needed for the inspection, such as planking, scaffolding, bosun's chairs, and portable ladders, should be available if needed.

5.2 MODES OF DETERIORATION AND FAILURE

Contaminants in fluids handled in pressure vessels may react with metals and cause corrosion. Stress reversals in parts of equipment are common, particularly at points of high secondary stress. If stresses are high and reversals are frequent, failure of parts may occur because of fatigue. Fatigue failures in pressure vessels may also occur because of cyclic temperature and pressure changes. Locations where metals with different thermal coefficients of expansion are welded together may be susceptible to thermal fatigue.

Deterioration or creep may occur if equipment is subjected to temperatures above those for which it is designed. Since metals weaken at higher temperatures, such distortion may cause failures, particularly at points of stress concentration. If excessive temperatures are encountered, structural-property and chemical changes in metals may take place that may permanently weaken equipment. Creep is dependent on time, temperature, and stress, so the actual or estimated levels of these quantities shall be used in any evaluations.

At subfreezing temperatures, water and some chemicals handled in pressure vessels may freeze and cause failure.

At ambient temperatures, carbon, low-alloy, and other ferritic steels may be susceptible to brittle failure. A number of failures have been attributed to brittle fracture of steels that were exposed to temperatures below their transition temperature and to pressures greater than 20 percent of the required hydrostatic test pressure; most brittle fractures, however, have occurred on the first application of a particular stress level (the first hydrotest or overload). Although the potential for a brittle failure because of excessive operating conditions below the transition temperature shall be evaluated, the potential for a brittle failure because of rehydrotesting or pneumatic testing of equipment or the addition of any other additional loadings shall also be evaluated. Special attention should be given to low-alloy steels (especially 2¼ Cr-1Mo) because they may be prone to temper embrittlement. [*Temper embrittlement* is a loss of ductility and notch toughness due to postweld heat treatment or high-temperature service (above 700°F) (370°C).]

Other forms of deterioration, such as stress corrosion cracking, hydrogen attack, carburization, graphitization, and erosion, may also occur under special circumstances. These forms of deterioration are more fully discussed in Chapter II of the API *Guide for Inspection for Refinery Equipment.*

5.3 CORROSION-RATE DETERMINATION

For a new vessel or for a vessel for which service conditions are being changed, one of the following methods shall be employed to determine the vessel's probable corrosion rate. The remaining wall thickness at the time of the next inspection can be estimated from this rate.

a. A corrosion rate may be calculated from data collected by the owner or user on vessels providing the same or similar service.
b. If data on vessels providing the same or similar service are not available, a corrosion rate may be estimated from the owner's or user's experience or from published data on vessels providing comparable service.
c. If the probable corrosion rate cannot be determined by either item a or item b above, on-stream determinations shall be made after approximately 1000 hours of service by using suitable corrosion monitoring devices or actual nondestructive thickness measurements of the vessel or system. Subsequent determinations shall be made after appropriate intervals until the corrosion rate is established.

If it is determined that an inaccurate corrosion rate has been assumed, the rate to be used for the next period shall be increased or may be decreased to agree with the actual rate.

5.4 MAXIMUM ALLOWABLE WORKING PRESSURE DETERMINATION

The maximum allowable working pressure for the continued use of a pressure vessel shall be based on computations that are determined using the latest edition of the ASME Code or the construction code to which the vessel was built. The resulting maximum allowable working pressure from these computations shall not be greater than the original max-

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imum allowable working pressure unless a rerating is performed in accordance with 7.3.

Computations may be made only if the following essential details comply with the applicable requirements of the code being used: head, shell, and nozzle reinforcement designs; material specifications; allowable stresses; weld efficiencies; inspection acceptance criteria; and cyclical service requirements. In corrosive service, the wall thickness used in these computations shall be the actual thickness as determined by inspection (see 5.7) but shall not be greater than the thickness reported in the material test report or the manufacturer's data report, where available, minus twice the estimated corrosion loss before the date of the next inspection, except as modified in 6.3. Allowance shall be made for the other loadings in accordance with the applicable provisions of the ASME Code.

5.5 DEFECT INSPECTION

Vessels shall be examined for visual indications of distortion. If any distortion of a vessel is suspected or observed, the overall dimensions of the vessel shall be checked to confirm whether or not the vessel is distorted and, if it is distorted, to determine the extent and seriousness of the distortion. The parts of the vessel that should be inspected most carefully depend on the type of vessel and its operating conditions. The authorized pressure vessel inspector should be familiar with the operating conditions of the vessel and with the causes and characteristics of potential defects and deterioration. (For recommended inspection practices for pressure vessels, see API Recommended Practice 572.)

Careful visual examination is the most important and the most universally accepted method of inspection. Other methods that may be used to supplement visual inspection include (a) magnetic-particle examination for cracks and other elongated discontinuities in magnetic materials; (b) fluorescent or dye-penetrant examination for disclosing cracks, porosity, or pin holes that extend to the surface of the material and for outlining other surface imperfections, especially in nonmagnetic materials; (c) radiographic examination; (d) ultrasonic thickness measurement and flaw detection; (e) eddy current examination; (f) metallographic examination; (g) acoustic emission testing; hammer testing while not under pressure; and (h) pressure testing. (Section V of the ASME Code can be used as a guide for many of the nondestructive examination techniques.)

Adequate surface preparation is important for proper visual examination and for the satisfactory application of any auxiliary procedures, such as those mentioned above. The type of surface preparations required depends on the individual circumstances, but surface preparations such as wire brushing, blasting, chipping, grinding, or a combination of these preparations may be required.

If external or internal coverings, such as insulation, refractory protective linings, and corrosion-resistant linings, are in good condition and there is no reason to suspect that an unsafe condition is behind them, it is not necessary to remove them for inspection of the vessel; however, it may be advisable to remove small portions of the coverings to investigate their condition and effectiveness and the condition of the metal underneath them.

Where operating deposits, such as coke, are normally permitted to remain on a vessel surface, it is particularly important to determine whether such deposits adequately protect the vessel surface from deterioration. To determine this, spot examinations in which the deposit is thoroughly removed from selected critical areas may be required.

Where vessels are equipped with removable internals, the internals need not be removed completely as long as reasonable assurance exists that deterioration in regions rendered inaccessible by the internals is not occurring to an extent beyond that found in more accessible parts of the vessel.

5.6 INSPECTION OF PARTS

The following inspections are not all inclusive for every vessel, but they do include the features that are common to most vessels and that are most important. Authorized pressure vessel inspectors must supplement this list with any additional items necessary for the particular vessel or vessels involved.

a. Examine the surfaces of shells and heads carefully for possible cracks, blisters, bulges, and other signs of deterioration. Pay particular attention to the skirt and to support-attachment and knuckle regions of the heads. If evidence of distortion is found, it may be necessary to make a detailed check of the actual contours or principal dimensions of the vessel and to compare those contours and dimensions with the original design details.

b. Examine welded joints and the adjacent heat-affected zones for service-induced cracks or other defects. On riveted vessels, examine rivet head, butt strap, plate, and caulked edge conditions. If rivet-shank corrosion is suspected, hammer testing or spot radiography at an angle to the shank axis may be useful.

c. Examine the surfaces of all manways, nozzles, and other openings for distortion, cracks, and other defects, paying particular attention to the welding used to attach the parts and their reinforcements. Normally, weep holes in reinforcing plates should remain open to provide visual evidence of leakage as well as to prevent pressure build-up in the cavity. Examine accessible flange faces for distortion and determine the condition of gasket-seating surfaces.

API Recommended Practice 574 can provide more information on the inspection of piping, valves, and fittings associated with pressure vessels. API Recommended Practice 572 can provide more information on pressure vessel inspection.

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5.7 CORROSION AND MINIMUM THICKNESS EVALUATION

Corrosion may cause a uniform loss (a general, relatively even wastage of a surface area) or may cause a pitted appearance (an obvious, irregular surface wastage). Uniform corrosion may be difficult to detect visually, and thickness readings may be necessary to determine its extent. Pitted surfaces may be thinner than they appear visually, and when there is uncertainty about the original surface location, thickness determinations may also be necessary.

The minimum actual thickness and maximum corrosion rate for any part of a vessel may be adjusted at any inspection. When the minimum actual thickness or maximum corrosion rate is to be adjusted, one of the following should be considered:

a. Any suitable nondestructive examination, such as ultrasonic or radiographic examination, that will not affect the safety of the vessel may be used as long as it will provide minimum thickness determinations. When a measurement method produces considerable uncertainty, test holes may be drilled, or other nondestructive techniques, such as ultrasonic A-scan, B-scan, or C-scan, may be employed. Profile radiography may be also employed.

b. If suitable openings are available, measurements may be taken through them.

c. The depth of corrosion may be determined by gauging the uncorroded surfaces within the vessel when such surfaces are in the vicinity of the corroded area.

d. For a corroded area of considerable size in which the circumferential stresses govern, the least thickness along the most critical element of the area may be averaged over a length not exceeding the following:

1. For vessels with inside diameters less than or equal to 60 inches (150 centimeters), one-half the vessel diameter or 20 inches (50 centimeters), whichever is less.
2. For vessels with inside diameters greater than 60 inches (150 centimeters), one-third the vessel diameter or 40 inches (100 centimeters), whichever is less.

When the area contains an opening, the distance on either side of the opening within which the thicknesses are averaged shall not extend beyond the limits of the reinforcement as defined in the ASME Code. If, because of wind loads or other factors, the longitudinal stresses govern, the least thickness in a similarly determined length of arc in the most critical plane perpendicular to the axis of the vessel also shall be averaged for computation of the longitudinal stresses. The thickness used for determining corrosion rates at the respective locations shall be the average thickness determined as in the preceding. For the purposes of 5.4, the actual thickness as determined by inspection shall be understood to mean the most critical value of the average thickness that has been determined.

e. Widely scattered pits may be ignored as long as the following are true:

1. No pit depth is greater than one-half the vessel wall thickness exclusive of the corrosion allowance.
2. The total area of the pits does not exceed 7 square inches (45 square centimeters) within any 8-inch (20-centimeter) diameter circle.
3. The sum of their dimensions along any straight line within the circle does not exceed 2 inches (5 centimeters).

f. As an alternative to the procedures just described, any components with thinning walls that, because of corrosion or other wastage, are below the minimum required wall thicknesses may be evaluated to determine if they are adequate for continued service. The thinning components may be evaluated by employing the design by analysis methods of Section VIII, Division 2, Appendix 4, of the ASME Code. These methods may also be used to evaluate blend ground areas where defects have been removed. It is important to ensure that there are no sharp corners in blend ground areas to minimize stress concentration effects.

When using this criteria, the stress value used in the original pressure vessel design shall be substituted for the S_m value of Division 2 if the design stress is less than or equal to ⅔-specified minimum yield strength (SMYS) at temperature. If the original design stress is greater than ⅔-specified minimum yield strength at temperature, then ⅔-specified minimum yield strength shall be substituted for S_m. When this approach is to be used, consulting with a pressure vessel engineer experienced in pressure vessel design is required.

g. When the surface at a weld with a joint factor of other than 1.0, as well as surfaces remote from the weld, is corroded, an independent calculation using the appropriate weld joint factor must be made to determine if the thickness at the weld or remote from the weld governs the allowable working pressure. For this calculation, the surface at a weld includes 1 inch (2.5 centimeters) on either side of the weld or twice the minimum thickness on either side of the weld, whichever is greater.

h. When measuring the corroded thickness of ellipsoidal and torispherical heads, the governing thickness may be as follows:

1. The thickness of the knuckle region with the head rating calculated by the appropriate head formula.
2. The thickness of the central portion of the dished region, in which case the dished region may be considered a spherical segment whose allowable pressure is calculated by the code formula for spherical shells.

The spherical segment of both ellipsoidal and torispherical heads shall be considered to be that area located entirely within a circle whose center coincides with the center of the head and whose diameter is equal to 80 percent of the shell diameter. The radius of the dish of torispherical heads is to be used as the radius of the spherical segment (equal to the diameter of the shell for standard heads, though other radii have been permitted). The radius of the spherical segment of ellipsoidal heads shall be considered to be the equivalent spherical radius K_1D, where D is the shell diameter (equal to the major

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axis) and K_1 is given in Table 1. In Table 1, h is one-half the length of the minor axis [equal to the inside depth of the ellipsoidal head measured from the tangent line (headbend line)]. For many ellipsoidal heads, $D/2h$ equals 2.0.

Table 1—Values of Spherical Radius Factor K_1

$D/2h$	K_1
3.0	1.36
2.8	1.27
2.6	1.18
2.4	1.08
2.2	0.99
2.0	0.90
1.8	0.81
1.6	0.73
1.4	0.65
1.2	0.57
1.0	0.50

Note: The equivalent spherical radius equals K_1D; the axis ratio equals $D/2h$. Interpolation is permitted for intermediate values.

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6 Inspection and Testing of Pressure Vessels and Pressure-Relieving Devices

6.1 GENERAL

Pressure vessels shall be inspected at the time of installation. Internal field inspections of new vessels are not required as long as a manufacturer's data report assuring that the vessels are satisfactory for their intended service is available. To ensure vessel integrity, all pressure vessels shall be inspected at the frequencies provided in this section.

In selecting the technique(s) to be used for the inspection of a pressure vessel, both the condition of the vessel and the environment in which it operates should be taken into consideration. The inspection, as deemed necessary by the authorized pressure vessel inspector, may include many of a num-ber of nondestructive techniques, including visual inspection. Internal inspection is preferred because process side degradation (corrosion, erosion, and environmental cracking) can be nonuniform throughout the vessel and, therefore, difficult to locate by external NDE. On-stream inspection may be acceptable in lieu of internal inspection for vessels under the specific circumstances defined in 6.3. In situations where on-stream inspection is acceptable, such inspection may be conducted either while the vessel is out of service and depressurized or on stream and under pressure. Except in response to an apparent need, such as when environmental cracking (see *Guide for Inspection of Refinery Equipment,* Chapter II) is suspected, inspection techniques exceeding the examination requirements used in the design and fabrication of the vessel are not required.

The appropriate inspection must provide the information necessary to determine that all of the essential sections or components of the vessel are safe to operate until the next scheduled inspection. The risks associated with operational shutdown and start-up and the possibility of increased corrosion due to exposure of vessel surfaces to air and moisture should be evaluated when an internal inspection is being planned.

6.2 EXTERNAL INSPECTION

Each vessel aboveground shall be given a visual external inspection, preferably while in operation, at least every 5 years or at the same interval as the required internal or on-stream inspection, whichever is less. The inspection shall, at the least, determine the condition of the exterior insulation, the condition of the supports, the allowance for expansion, and the general alignment of the vessel on its supports. Any signs of leakage should be investigated so that the sources can be established. Inspection for corrosion under insulation (CUI) shall be considered for externally-insulated vessels subject to moisture ingress and that operate between –4°C and 120°C, or are in intermittent service. This inspection may require removal of some insulation. It is not normally necessary to remove insulation if the entire vessel shell is always operated at a temperature sufficiently low (below –4°C) or sufficiently high (above 120°C) to prevent the presence or condensation of moisture under the insulation. Alternatively, shell thickness measurements done internally at typical problem areas (for example, stiffening rings, around nozzles, and other locations which tend to trap moisture or allow moisture ingress) may be performed during internal inspections.

Buried vessels shall be inspected to determine their external environmental condition. The inspection interval shall be based on corrosion-rate information obtained from one or more of the following methods: (a) during maintenance activity on adjacent connecting piping of similar material; (b) from the interval examination (specified in the paragraph above) of similarly buried corrosion test coupons of similar material; (c) from representative portions of the actual vessel; or (d) from a vessel in similar circumstances.

Vessels that are known to have a remaining life of over 10 years or that are protected against external corrosion—for example, (a) vessels insulated effectively to preclude the entrance of moisture, (b) jacketed cryogenic vessels, (c) vessels installed in a cold box in which the atmosphere is purged with an inert gas, and (d) vessels in which the temperature being maintained is sufficiently low or sufficiently high to preclude the presence of water—do not need to have insulation removed for the external inspection. However, the condition of their insulating system or their outer jacketing, such as the cold box shell, shall be observed at least every 5 years and repaired if necessary.

6.3 INTERNAL AND ON-STREAM INSPECTION

The period between internal or on-stream inspections shall not exceed one half the estimated remaining life of the vessel based on corrosion rate or 10 years, whichever is less. In cases where the remaining safe operating life is estimated to be less than 4 years, the inspection interval may be the full remaining safe operating life up to a maximum of 2 years.

For pressure vessels that are in noncontinuous service and are isolated from the process fluids such that they are not exposed to corrosive environments (such as inert gas purged or filled with noncorrosive hydrocarbons), the 10 years shall be the 10 years of actual service exposed life. Equipment that is not adequately protected from corrosive environments may experience significant internal corrosion while idle and should be carefully reviewed when setting inspection intervals. In no case should these exceed one-half the estimated remaining corrosion-rate life, or 10 years since the last inspection. External inspections for vessels in noncontinuous service remain the same as for continuous service, as outlined in 6.2.

Except as noted below, internal inspection is normally the preferred method of inspection and shall be conducted on vessels subject to significant localized corrosion and other

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types of damage. At the discretion of the authorized pressure vessel inspector, on-stream inspection may be substituted for internal inspection in the following situations:

a. When size, configuration, or lack of access makes vessel entry for internal inspection physically impossible.
b. When the general corrosion rate of a vessel is known to be less than 0.005 inch (0.125 millimeter) per year and the estimated remaining life is greater than 10 years, and all of the following conditions are met:
   1. The corrosive character of the contents, including the effect of trace components, has been established by at least 5 years of the same or comparable service experience with the type of contents being handled.
   2. No questionable condition is disclosed by the external inspection specified in 6.2.
   3. The operating temperature of the steel vessel shell does not exceed the lower temperature limits for the creep-rupture range of the vessel material.
   4. The vessel is not considered to be subject to environmental cracking or hydrogen damage from the fluid being handled.
   5. The vessel is not strip-lined or plate-lined.

If the requirements of item b above are not met, as a result of conditions noted during the scheduled on-stream inspection, the next scheduled inspection shall be an internal inspection.

When an on-stream inspection is conducted in lieu of an internal inspection, a thorough examination shall be performed using ultrasonic thickness measurements, or radiography, or other appropriate means of NDE to measure metal thicknesses and/or assess the integrity of the metal and welds. If an on-stream inspection is conducted, the authorized pressure vessel inspector shall be given sufficient access to all parts of the vessel (heads, shell, and nozzles) so that the inspector is satisfied that an accurate assessment of the vessel condition can be made.

A representative number of thickness measurements must be conducted on each vessel to satisfy the requirements for an internal or on-stream inspection. For example, the thickness for all major components (shells, heads, cone sections) and a representative sample of vessel nozzles should be measured and recorded, and the remaining life and next inspection interval should be calculated for the limiting component. A decision on the number and location of the thickness measurements should consider results from previous inspections, if available, and the potential consequence of loss of containment. Measurements at a number of thickness measurement locations (TMLs) are intended to establish general and localized corrosion rates in different sections of the vessel. A minimal number of TMLs are acceptable when the established rate of corrosion is low and not localized. For pressure vessels susceptible to localized corrosion, it is vital that those knowledgeable in localized corrosion mechanisms be consulted about the appropriate placement and number of TMLs. Additionally, for localized corrosion, it is important that inspections are conducted using scanning methods such as profile radiography, scanning ultrasonics, and/or other suitable NDE methods that will reveal the scope and extent of localized corrosion.

The remaining life of the vessel shall be calculated from the following formula:

$$\text{Remaining life (years)} = \frac{t_{\text{actual}} - t_{\text{minimum}}}{\text{corrosion rate}}$$

[inches (millimeters) per year]

Where:

t_{actual} = the thickness, in inches (millimeters), recorded at the time of inspection for a given location or component.

t_{minimum} = minimum allowable thickness, in inches (millimeters), for a given location or component.

$$\text{Corrosion rate} = \frac{t_{\text{previous}} - t_{\text{actual}}}{\text{years between } t_{\text{previous}} \text{ and } t_{\text{actual}}}$$

t_{previous} = the thickness, in inches (millimeters), at the same location as t_{actual} measured during a previous inspection.

A statistical analysis may be used in the corrosion rate and remaining life calculations for the pressure vessel sections. This statistical approach may be applied for assessment of substituting an internal inspection (item b in the preceding), or for determining the internal inspection interval. Care must be taken to ensure that the statistical treatment of data results reflects the actual condition of the vessel section. Statistical analysis is not applicable to vessels with significant localized corrosion.

The determination of corrosion rate may include thickness data collected at more than two different times. Suitable use of short-term versus long-term corrosion rates shall be determined by the authorized pressure vessel inspector. When there is a discrepancy between short-term and long-term corrosion rates, a pressure vessel engineer experienced in corrosion may need to be consulted about the use of these rates, at the discretion of the inspector, for calculating the remaining life and next inspection date.

For a large vessel with two or more zones of differing corrosion rates, each zone may be treated independently regarding the interval between inspections or for substituting the internal inspection with an on-stream inspection. If a multizone analysis is used, the zone with the shortest remaining half-life shall be used as the limiting case for setting the internal inspection interval or for substituting the internal inspection with an on-stream inspection.

An alternative method to establish the required inspection interval based on remaining life is by calculation of the projected maximum allowable working pressure (MAWP) of

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each vessel component as described in 5.4. This procedure may be iterative involving selection of an inspection interval, determination of the corrosion loss expected over the interval, and calculation of the projected MAWP. The inspection interval is within the maximum permitted as long as the projected MAWP of the limiting component is not less than the lower of the name plate or rerated MAWP. The maximum inspection interval using this method is also 10 years.

When problems are experienced with external loading, faulty material, or fabrication the remaining life as determined above shall be reduced to recognize those conditions. If deterioration due to conditions such as those mentioned in 5.2 is detected, the inspection interval must be appropriately adjusted.

If the service conditions of a vessel are changed, the maximum operating pressure, the maximum and minimum operating temperature, and the period of operation until the next inspection shall be established for the new service conditions.

If both the ownership and the location of a vessel are changed, the vessel shall be internally and externally inspected before it is reused, and the allowable conditions of service and the next period of inspection shall be established for the new service.

6.4 PRESSURE TEST

When the authorized pressure vessel inspector believes that a pressure test is necessary or when, after certain repairs or alterations, the inspector believes that one is necessary (see 7.2.9), the test shall be conducted at a pressure in accordance with the construction code used for determining the maximum allowable working pressure. To minimize the risk of brittle fracture during the test, the metal temperature should be maintained at least 30°F (–1°C) above the minimum design metal temperature for vessels that are more than 2 inches thick, or 10°F (–12°C) above for vessels that have a thickness of 2 inches or less. The test temperature need not exceed 120°F (50°C) unless there is information on the brittle characteristics of the vessel material indicating that a lower test temperature is acceptable or a higher test temperature is needed.[5]

Pneumatic testing may be used when hydrostatic testing is impracticable because of temperature, foundation, refractory lining, or process reasons; however, the potential personnel and property risks of pneumatic testing shall be considered before such testing is carried out. As a minimum, the inspection precautions contained in the ASME Code shall be applied in any pneumatic testing. Before applying a hydrostatic test to equipment, consideration should be given to the supporting structure and the foundation design.

When a pressure test is to be conducted in which the test pressure will exceed the set pressure of the safety relief valve with the lowest setting, the safety relief valve or valves should be removed. An alternative to removing the safety relief valves is to use test clamps to hold down the valve disks. Applying an additional load to the valve spring by turning the compression screw is not recommended. Other appurtenances, such as gauge glasses, pressure gauges, and rupture disks, that may be incapable of withstanding the test pressure should also be removed or should be blanked off or vented. When the pressure test has been completed, pressure relief devices of the proper settings and other appurtenances removed or made inoperable during the pressure test shall be reinstalled or reactivated.

[5]For vessels without minimum design metal temperature, the minimum acceptable operating temperature should be used in lieu of the minimum design metal temperature.

6.5 PRESSURE-RELIEVING DEVICES

Pressure relief valves shall be tested and repaired by repair organizations experienced in valve maintenance. Each repair organization shall have a fully documented quality control system. As a minimum, the following requirements and pieces of documentation should be included in the quality control system:

a. Title page.
b. Revision log.
c. Contents page.
d. Statement of authority and responsibility.
e. Organizational chart.
f. Scope of work.
g. Drawings and specification controls.
h. Material and part control.
i. Repair and inspection program.
j. Welding, nondestructive examination, and heat treatment procedures.
k. Valve testing, setting, leak testing, and sealing.
l. General example of the valve repair nameplate.
m. Procedures for calibrating measurement and test gauges.
n. Controlled copies of the manual.
o. Sample forms.
p. Repair personnel training or qualifications.

Each repair organization shall also have a fully documented training program that shall ensure that repair personnel are qualified within the scope of the repairs.

Pressure relief valves shall be tested at intervals that are frequent enough to verify that the valves perform reliably. This may include testing pressure relief valves on newly installed equipment. Pressure-relieving devices should be tested and maintained in accordance with API Recommended Practice 576, which replaces Chapter XVI of the API *Guide for Inspection of Refinery Equipment*. Other pressure-relieving devices, such as rupture disks and vacuum-breaker valves, shall be thoroughly examined at intervals determined on the basis of service.

The intervals between pressure-relieving-device testing or inspection should be determined by the performance of the

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devices in the particular service concerned. Test or inspection intervals on pressure-relieving devices in typical process services should not exceed 5 years, unless service experience indicates that a longer interval is acceptable. For clean (nonfouling), noncorrosive services, maximum intervals may be increased to 10 years. When service records indicate that a pressure-relieving device was heavily fouled or stuck in the last inspection or test, the service interval shall be reduced if the review shows that the device may not perform reliably in the future. The review should include an effort to determine the cause of the fouling or the reasons for the relief device not operating properly.

6.6 RECORDS

Pressure vessel owners and users shall maintain permanent and progressive records of their pressure vessels. Permanent records will be maintained throughout the service life of each vessel; progressive records will be regularly updated to include new information pertinent to the operation, inspection, and maintenance history of the vessel.

Pressure vessel records shall contain three types of vessel information pertinent to mechanical integrity as follows:

a. Construction and design information. For example, equipment serial number or other identifier, manufacturers' data reports (MDRs), design specification data, design calculations (where MDRs are unavailable), and construction drawings.

b. Operating and inspection history. For example, operating conditions, including process upsets that may affect mechanical integrity, inspection reports, and data for each type of inspection conducted (for example, internal, external, thickness measurements), and inspection recommendations for repair. See Appendix C for sample pressure vessel inspection records. Inspection reports shall document the date of each inspection and/or test, the date of the next scheduled inspection, the name of the person who performed the inspection and/or test, the serial number or other identifier of the equipment inspected, a description of the inspection and/or test performed, and the results of the inspection and/or test.

c. Repair, alteration, and rerating information. For example, (1) repair and alteration forms like that shown in Appendix D, (2) reports indicating that equipment still in-service with identified deficiencies or recommendations for repair are suitable for continued service until repairs can be completed, and (3) rerating documentation (including rerating calculations, new design conditions, and evidence of stamping).

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7 Repairs, Alterations, and Rerating of Pressure Vessels

7.1 GENERAL

This section covers repairs and alterations to pressure vessels by welding. The requirements that must be met before pressure vessels can be rerated are also covered in this section. When repairs or alterations have to be performed, the applicable requirements of the ASME Code, the codes to which the vessels were built, or other specific pressure vessel rating codes shall be followed. Before any repairs or alterations are performed, all proposed methods of execution, all materials, and all welding procedures that are to be used must be approved by the authorized pressure vessel inspector and, if necessary, by a pressure vessel engineer experienced in pressure vessel design, fabrication, or inspection.

7.1.1 Authorization

All repair and alteration work must be authorized by the authorized pressure vessel inspector before the work is started by a repair organization (see 3.13). Authorization for alterations to pressure vessels that comply with Section VIII, Divisions 1 and 2, of the ASME Code and for repairs to pressure vessels that comply with Section VIII, Division 2, of the ASME Code may not be given until a pressure vessel engineer experienced in pressure vessel design has been consulted about the alterations and repairs and has approved them. The authorized pressure vessel inspector will designate the fabrication approvals that are required. The authorized pressure vessel inspector may give prior general authorization for limited or routine repairs as long as the inspector is sure that the repairs are the kind that will not require pressure tests.

7.1.2 Approval

The authorized pressure vessel inspector shall approve all specified repair and alteration work after an inspection of the work has proven the work to be satisfactory and any required pressure test has been witnessed.

7.1.3 Defect Repairs

A crack in a welded joint and a defect in a plate may be repaired by preparing a U- or V-shaped groove to the full depth and length of the crack and then filling the groove with weld metal deposited in accordance with 7.2. No crack shall be repaired without authorization from the authorized pressure vessel inspector. Repairing a crack at a discontinuity, where stress concentrations may be serious, should not be attempted without prior consultation with a pressure vessel engineer experienced in pressure vessel design.

Corroded areas, as defined by 5.7, may be restored with weld metal deposited in accordance with 7.2. Surface irregularities and contamination shall be removed before welding. The nondestructive examination and inspection appropriate for the extent of restoration being performed shall be specified in the repair procedure.

7.2 WELDING

All repair and alteration welding shall be in accordance with the applicable requirements of the ASME Code, except as permitted in 7.2.11.

7.2.1 Procedures and Qualifications

The repair organization shall use qualified welders and welding procedures qualified in accordance with the applicable requirements of Section IX of the ASME Code.

7.2.2 Qualification Records

The repair organization shall maintain records of its qualified welding procedures and its welding performance qualifications. These records shall be available to the inspector prior to the start of welding. The repair organization's qualified welding procedures and welding performance qualifications shall be in accord with the appropriate code.

7.2.3 Heat Treatment—Preheating

Note: Before preheating is used, a metallurgical review must be conducted to determine if the vessel was postweld heat treated due to the characteristics of the fluid contained in it.

For alterations or repairs of vessels initially postweld heat treated as a code requirement and constructed of P-1 and P-3 steels listed in the ASME Code, preheating to not less than 300°F (150°C) may be considered as an alternative to postweld heat treatment. The use of this minimum preheat alternative is to be restricted to preheating those steels that meet the exemption criteria found in UCS-56(f) (1) through (4) of Section VIII, Division 1, of the ASME Code. If the depth of the weld exceeds the maximum thickness exempt from postweld heat treatment, the repair weld or alteration shall be postweld heat treated in accordance with the applicable requirements of the ASME Code.

Vessels constructed of other steels that initially required postweld heat treatment shall normally be postweld heat treated if alterations or repairs involving strength welding are performed. When the preheat or temper-bead alternative is used as an alternative to postweld heat treatment, the postweld heat treatment joint efficiency factor may be continued if the factor has been used in the currently rated design (see note). Consulting with a pressure vessel engineer experienced in pressure vessel design is required if the preheat or temper-bead alternative is desired.

7.2.4 Temper-Bead Welding

Note: Before temper-bead welding is used, a metallurgical review must be conducted to determine if the vessel was postweld heat-treated due to the characteristics of the fluid contained in it.

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For weld repairs of vessels originally postweld heat treated as a code requirement and constructed of P-1 and P-3 steels listed in the ASME Code, a temper-bead welding technique may be used in lieu of postweld heat treatment. If a temper-bead welding technique is to be used, the following requirements must be met:

a. The weld area shall be preheated and maintained at a minimum temperature of 350°F (175°C) during welding. The maximum interpass temperature shall be 450°F (230°C).
b. The initial layer of weld metal shall be deposited over the entire area with ⅛-inch (3-millimeter) maximum diameter electrodes. Approximately one-half the thickness of this layer shall be removed by grinding before subsequent layers are deposited. Subsequent layers shall be deposited with 5/32-inch (4-millimeter) maximum diameter electrodes in a manner to ensure tempering of the prior beads and their heat-affected zones. The final temper-bead reinforcement layer shall be removed substantially flush with the surface of the base material or the previous weld layer.
c. Heat input shall be controlled within a specified range of welding current and voltage.
d. The weld area shall be maintained at a temperature of 500°F ±50°F (260°C ±28°C) for a minimum of 2 hours after completion of the weld repair.
e. The repair welding shall be witnessed by the authorized pressure vessel inspector.
f. The weld metal shall be deposited by the manual shielded metal arc process using low-hydrogen electrodes. The maximum bead width shall be four times the electrode core diameter.
g. The use of the temper-bead welding technique is to be restricted to temper-bead welding those steels that meet the postweld heat treatment exemption criteria found in UCS-56(f) (1) through (4) of Section VIII, Division 1, of the ASME Code. If the depth of the repair exceeds the maximum thickness exempt from local postweld heat treatment, the repair weld shall be postweld heat-treated in accordance with the applicable requirements of the ASME Code.

7.2.5 Local Postweld Heat Treatment

Note: Before local postweld heat treatment is used, a metallurgical review must be conducted to determine if the vessel was postweld heat treated due to the characteristics of the fluid contained in it.

Local postweld heat treatment (PWHT) may be substituted for 360-degree banding on local repairs on all materials, provided that the following precautions are taken and requirements are met:

a. The application is reviewed, and a procedure is developed by pressure vessel engineers experienced in the appropriate engineering specialties.
b. The suitability of the procedure is evaluated. In evaluating the suitability of the procedure, the following shall be considered: applicable factors, such as base metal thickness, decay thermal gradients, and material properties (hardness, constituents, strength, and the like); changes due to local postweld heat treatment; the need for full penetration welds; and surface and volumetric examinations after local postweld heat treatment. In evaluating and developing local postweld heat treatment procedures, the overall and local strains and distortions resulting from the heating of a local restrained area of the pressure vessel shell shall be considered.
c. A preheat of 300°F (150°C) or higher, as specified by specific welding procedures, is maintained during welding.
d. The required local postweld heat treatment temperature shall be maintained for a distance of not less than two times the base metal thickness measured from the weld. The local postweld heat treatment temperature shall be monitored by a suitable number of thermocouples (at least two). (When determining the number of thermocouples necessary, the size and shape of the area being heat treated should be considered.)
e. Heat shall be applied to any nozzle or any attachment within the local postweld heat treatment area.

7.2.6 Repairs to Stainless Steel Weld Overlay and Cladding

The repair procedure(s) to restore removed, corroded, or missing clad or overlay areas shall be reviewed and endorsed prior to implementation by the pressure vessel engineer and authorized by the inspector.

Consideration shall be given to factors which may augment the repair sequence such as stress level, P number of base material, service environment, possible previously dissolved hydrogen, type of lining, deterioration of base metal properties (by temper embrittlement of chromium-molybdenum alloys), minimum pressurization temperatures, and a need for future periodic examination.

For equipment which is in hydrogen service at an elevated temperature or which has exposed base metal areas open to corrosion which could result in a significant atomic hydrogen migration in the base metal, the repair must additionally be considered by the pressure vessel engineer for factors affecting the following:

a. Outgassing base metal.
b. Hardening of base metal due to welding, grinding, or arc gouging.
c. Preheat and interpass temperature control.
d. Postweld heat treatment to reduce hardness and restore mechanical properties.

Repairs shall be monitored by an inspector to assure compliance to repair requirements. After cooling to ambient temperatures, the repair shall be inspected by the liquid penetrant method, according to ASME Code, Section VIII, Division 1, Appendix 8.

For vessels constructed with P-3, P-4, or P-5 base materials, the base metal in the area of repair should be examined for cracking by the ultrasonic examination in accordance with

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ASME Code, Section V, Article 5, paragraph T-543. This inspection is most appropriately accomplished following a delay of at least 24 hours after completed repairs for equipment in hydrogen service and for chromium-molybdenum alloys that could be affected by delayed cracking.

7.2.7 Design

Butt joints shall have complete penetration and fusion. Parts should be replaced when repairing them is likely to be inadequate. Part replacements shall be fabricated according to the applicable requirements of the appropriate code. New connections may be installed on vessels as long as the design, location, and method of attachment comply with the applicable requirements of the appropriate code.

Fillet-welded patches require special design considerations, especially relating to efficiency. Fillet-welded patches may be used to make temporary repairs, and the use of fillet-welded patches may be subject to the patches' acceptance in the jurisdiction in which they are required. Temporary repairs using fillet-welded patches shall be approved by a pressure vessel engineer competent in pressure vessel design; and the temporary repairs should be removed and replaced with suitable permanent repairs at the next available maintenance opportunity. Temporary repairs may remain in place for a longer period of time only if evaluated, approved, and documented by the pressure vessel engineer and the authorized API pressure vessel inspector. Fillet-welded patches may be applied to the internal or external surfaces of shells, heads, and headers as long as, in the judgment of the authorized pressure vessel inspector, either of the following is true:

a. The fillet-welded patches provide design safety equivalent to reinforced openings designed according to the applicable section of the ASME Code.
b. The fillet-welded patches are designed to absorb the membrane strain of the parts so that in accordance with the rules of the applicable section of the ASME Code, the following result:
   1. The allowable membrane stress is not exceeded in the vessel parts or the patches.
   2. The strain in the patches does not result in fillet-weld stresses that exceed allowable stresses for such welds.

Overlay patches shall have rounded corners. Flush (insert) patches shall also have rounded corners, and they shall be installed with full-penetration butt joints.

7.2.8 Material

The material used in making repairs or alterations shall conform to the applicable section of the ASME Code. The material shall be of known weldable quality and be compatible with the original material. Carbon or alloy steel with a carbon content over 0.35 percent shall not be welded.

7.2.9 Inspection

Acceptance criteria for a welded repair or alteration should include nondestructive examination techniques that are in accordance with the applicable sections of the ASME Code or another applicable vessel rating code. Where use of these nondestructive examination techniques is not possible or practical, alternative nondestructive examination methods may be used.

7.2.10 Testing

After repairs are completed, a pressure test shall be applied if the authorized pressure vessel inspector believes that one is necessary. A pressure test is normally required after an alteration. Subject to the approval of the jurisdiction (where the jurisdiction's approval is required), appropriate nondestructive examinations shall be required where a pressure test is not performed. Substituting nondestructive examination procedures for a pressure test after an alteration may be done only after a pressure vessel engineer experienced in pressure vessel design and the authorized pressure vessel inspector have been consulted.

7.2.11 Filler Metal

The filler metal used for weld repairs should have minimum specified tensile strength equal to or greater than the minimum specified tensile strength of the base metal. If a filler metal is used that has a minimum specified tensile strength lower than the minimum specified tensile strength of the base metal, the compatibility of the filler metal chemistry with the base metal chemistry shall be considered regarding weldability and service degradation. In addition, the following shall be met:

a. The repair thickness shall not be more than 50 percent of the required base metal thickness, excluding corrosion allowance.
b. The thickness of the repair weld shall be increased by a ratio of minimum specified tensile strength of the base metal and minimum specified tensile of the filler metal used for the repair.
c. The increased thickness of the repair shall have rounded corners and shall be blended into the base metal using a 3-to-1 taper.
d. The repair shall be made with a minimum of two passes.

7.3 RERATING

Rerating a pressure vessel by changing its temperature ratings or its maximum allowable working pressure may be done only after all of the following requirements have been met:

a. Calculations from either the manufacturer or an owner-user pressure vessel engineer (or his designated representative) experienced in pressure vessel design, fabrication, or inspection shall justify rerating.

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b. A rerating shall be established in accordance with the requirements of the construction code to which the pressure vessel was built or by computations that are determined using the appropriate formulas in the latest edition of the ASME Code if all of the essential details comply with the applicable requirements of the code being used.

c. Current inspection records verify that the pressure vessel is satisfactory for the proposed service conditions and that the corrosion allowance provided is appropriate. An increase in allowable working pressure or temperature shall be based on thickness data obtained from a recent internal or on-stream inspection.

d. The pressure vessel has at some time been pressure tested in accordance with the new service conditions, or the vessel integrity is maintained by special nondestructive evaluation inspection techniques in lieu of testing.

e. The pressure vessel inspection and rerating is acceptable to the authorized pressure vessel inspector.

The pressure vessel rerating will be considered complete when the authorized pressure vessel inspector oversees the attachment of an additional nameplate or additional stamping that carries the following information:

Rerated by _____
Maximum Allowable Working Pressure _____ psi at ____ F°
Date _____

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8 Alternative Rules for Exploration and Production Pressure Vessels

8.1 SCOPE AND SPECIFIC EXEMPTIONS

This section sets forth the minimum alternative inspection rules for pressure vessels that are exempt from the rules set forth in Section 6 except as referenced in paragraphs 8.4 and 8.5. Except for Section 6, all of the sections in this inspection code are applicable to Exploration and Production (E&P) pressure vessels. These rules are provided because of the vastly different characteristics and needs of pressure vessels used for E&P service. Typical E&P services are vessels associated with drilling, production, gathering, transportation, and treatment of liquid petroleum, natural gas, natural gas liquids, and associated salt water (brine).

The following are specific exemptions:

a. Portable pressure vessels and portable compressed gas containers associated with construction machinery, pile drivers, drilling rigs, well-servicing rigs and equipment, compressors, trucks, ships, boats, and barges shall be treated, for inspection and recording purposes, as a part of that machinery and shall be subject to prevailing rules and regulations applicable to that specific type of machine or container.

b. Pressure vessels referenced in Appendix A are exempt from the specific requirements of this inspection code.

8.2 GLOSSARY OF TERMS

8.2.1 class of vessels: Pressure vessels used in a common circumstance of service, pressure, and risk.

8.2.2 inspection: The external, internal, or on-stream evaluation (or any combination of the three) of a pressure vessel's condition.

a. **external inspection:** Evaluation performed from the outside of a pressure vessel using visual procedures to establish the suitability of the vessel for continued operation. The inspection may, or may not, be carried out while the vessel is in operation.

b. **internal inspection:** Evaluation performed from the inside of a pressure vessel using visual and/or nondestructive examination procedures to establish the suitability of the vessel for continued operation.

c. **on-stream inspection:** Evaluation performed from the outside of a pressure vessel using nondestructive examination procedures to establish the suitability of the vessel for continued operation. The vessel may, or may not, be in operation while the inspection is carried out.

d. **progressive inspection:** An inspection whose scope (coverage, interval, technique, and so forth) is increased as a result of inspection findings.

8.2.3 Section 8 vessel: A pressure vessel which is exempted from the rules set forth in Section 6 of this document.

8.3 INSPECTION PROGRAM

Each owner or user of Section 8 vessels shall have an inspection program that will assure that the vessels have sufficient integrity for the intended service. Each E&P owner or user shall have the option of employing, within the limitations of the jurisdiction in which the vessels are located, any appropriate engineering, inspection, classification, and recording systems which meet the requirements of this document.

8.3.1 On-Stream or Internal Inspections

a. Either an on-stream inspection or an internal inspection may be used interchangeably to satisfy inspection requirements. An internal inspection is required when the vessel integrity cannot be established with an on-stream inspection. When an on-stream inspection is used, a progressive inspection shall be employed.

b. In selecting the technique(s) to be utilized for the inspection of a pressure vessel, both the condition of the vessel and the environment in which it operates should be taken into consideration. The inspection may include any number of nondestructive techniques, including visual inspection, as deemed necessary by the owner-user.

c. At each on-stream or internal inspection, the remaining corrosion-rate life shall be determined as described in 8.3.2.

8.3.2 Remaining Corrosion Rate Life Determination:

For a new vessel, a vessel for which service conditions are being changed, or existing vessels, the remaining corrosion rate life shall be determined for each vessel or estimated for a class of vessels based on the following formula:

$$\text{Remaining life (years)} = \frac{t_{\text{actual}} - t_{\text{minimum}}}{\text{corrosion rate}}$$

[inches (millimeters) per year]

Where:

t_{actual} = the thickness, in inches (millimeters), measured at the time of inspection for a given location or component used to determine the minimum allowable thickness.

t_{minimum} = the minimum required thickness, excluding corrosion allowance.

t_{minimum} can be obtained by any one of the following:

a. The nominal thickness in the uncorroded condition, less the specified corrosion allowance.

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b. The original measured thickness, if documented, in the uncorroded condition, less the specified corrosion allowance.
c. Calculations in accordance with the requirements of the construction code to which the pressure vessel was built, or by computations that are determined using the appropriate formulas in the latest edition of the ASME Code, if all of the essential details comply with the applicable requirements of the code being used.

corrosion rate = loss of metal thickness, in inches (millimeters), per year. For vessels in which the corrosion rate is unknown, the corrosion rate shall be determined by one of the following methods:

1. A corrosion rate may be calculated from data collected by the owner or user on vessels in the same or similar service.
2. If data on vessels providing the same or similar service is not available, a corrosion rate may be estimated from the owner's or user's experience or from published data on vessels providing comparable service.
3. If the probable corrosion rate cannot be determined by either item a or item b above, on-stream determination shall be made after approximately 1000 hours of service by using suitable corrosion monitoring devices or actual nondestructive thickness measurements of the vessel or system. Subsequent determinations shall be made after appropriate intervals until the corrosion rate is established.

The remaining life shall be determined by an individual experienced in pressure vessel design and/or inspection. If it is determined that an inaccurate assumption has been made for either corrosion rate or thickness, the remaining life shall be increased or decreased to agree with the actual rate or thickness.

Other failure mechanisms (stress corrosion, brittle fracture, blistering, and so forth,) shall be taken into account in determining the remaining life of the vessel.

8.3.3 External Inspections

The following apply to external inspections:

a. The external visual inspection shall, at least, determine the condition of the shell, heads, nozzles, exterior insulation, supports and structural parts, pressure-relieving devices, allowance for expansion, and general alignment of the vessel on its supports. Any signs of leakage should be investigated so that the sources can be established. It is not necessary to remove insulation if the entire vessel shell is maintained at a temperature sufficiently low or sufficiently high to prevent the condensation of moisture. Refer to API Recommended Practice 572 for guidelines on external vessel inspections.
b. Buried sections of vessels shall be monitored to determine their external environmental condition. This monitoring shall be done at intervals that shall be established based on corrosion-rate information obtained during maintenance activity on adjacent connected piping of similar material, information from the interval examination of similarly buried corrosion test coupons of similar material, information from representative portions of the actual vessel, or information from a sample vessel in similar circumstances.
c. Vessels that are known to have a remaining life of over 10 years or that are protected against external corrosion—for example, (1) vessels insulated effectively to preclude the entrance of moisture, (2) jacketed cryogenic vessels, (3) vessels installed in a cold box in which the atmosphere is purged with an inert gas, and (4) vessels in which the temperature being maintained is sufficiently low or sufficiently high to preclude the presence of water—do not need to have insulation removed for the external inspection; however, the condition of their insulating system or their outer jacketing, such as the cold box shell, shall be observed at least every 5 years and repaired if necessary.

8.3.4 Vessel Classifications

The pressure vessel owner or user shall have the option to establish vessel inspection classes by grouping vessels into common classes of service, pressure, and/or risk. Vessel classifications shall be determined by an individual(s) experienced in the criteria outlined in the following. If vessels are grouped into classes (such as lower and/or higher risk), at a minimum, the following shall be considered to establish the risk class:

a. Potential for vessel failure, such as, minimum design metal temperature; potential for cracking, corrosion, and erosion; and the existence of mitigation factors.
b. Vessel history, design, and operating conditions, such as, the type and history of repairs or alterations, age of vessel, remaining corrosion allowance, properties of contained fluids, operating pressure, and temperature relative to design limits.
c. Consequences of vessel failure, such as, location of vessel relative to employees or the public, potential for equipment damage, and environmental consequences.

8.3.5 Inspection Intervals

The following apply to inspection intervals:

a. Inspections shall be performed at intervals determined by the vessel's risk classification. The inspection intervals for the two main risk classifications (lower and higher) are defined below. When additional classes are established, inspection and sampling intervals shall be set between the higher risk and lower risk classes as determined by the owner or user. If the owner or user decides to not classify vessels into risk classes, the inspection requirements and intervals of higher-risk vessels shall be followed.
b. Lower-risk vessels shall be inspected as follows:
1. Inspections on a representative sample of vessels in that class, or all vessels in that class, may be performed.

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2. External inspections shall be performed when an on-stream or internal inspection is performed or at shorter intervals at the owner or user's option.
3. On-stream or internal inspections shall be performed at least every 15 years or ¾-remaining corrosion-rate life, whichever is less.
4. Any signs of leakage or deterioration detected in the interval between inspections shall require an on-stream or internal inspection of that vessel and a reevaluation of the inspection interval for that vessel class.

c. Higher-risk vessels shall be inspected as follows:
1. External inpections shall be performed when an on-stream or internal inspection is performed or at shorter intervals at the owner or user's option.
2. On-stream or internal inspections shall be performed at least every 10 years or ½-remaining corrosion rate life, which ever is less.
3. In cases where the remaining life is estimated to be less than 4 years, the inspection interval may be the full remaining life up to a maximum of 2 years. Consideration should also be given to increasing the number of vessels inspected within that class to improve the likelihood of detecting the worst-case corrosion.
4. Any signs of leakage or deterioration detected in the interval between inspections shall require an on-stream or internal inspection of that vessel and a reevaluation of the inspection interval for that vessel class.

d. Pressure vessels (whether grouped into classes or not) shall be inspected at intervals sufficient to insure their fitness for continued service. Operational conditions and vessel integrity may require inspections at shorter intervals than the intervals stated above.

e. If service conditions change, the maximum operating temperature, pressure, and interval between inspections must be reevaluated.

f. For large vessels with two or more zones of differing corrosion rates, each zone may be treated independently regarding the interval between inspections.

8.3.6 Additional Inspection Requirements

Additional inspection requirements, regardless of vessel classification, exist for the following vessels:

a. Vessels that have changed ownership and location must have an on-stream or internal inspection performed to establish the next inspection interval and to assure that the vessel is suitable for its intended service. Inspection of new vessels is not required if a manufacturer's data report is available.

b. If a vessel is transferred to a new location, and it has been more than 5 years since the vessel's last inspection, an on-stream or internal inspection is required. (Vessels in truck-mounted, skid-mounted, ship-mounted, or barge-mounted equipment are not included.)

c. Air receivers (other than portable equipment) shall be inspected at least every 5 years.

d. Portable or temporary pressure vessels that are employed for the purpose of testing oil and gas wells during completion or recompletion shall be inspected at least once during each 3-year period of use. More frequent inspections shall be conducted if vessels have been in severe corrosive environments.

8.4 PRESSURE TEST

When a pressure test is conducted, the test shall be in accordance with the procedures in 6.4.

8.5 SAFETY RELIEF DEVICES

Safety relief devices shall be inspected, tested, and repaired in accordance with 6.5.

8.6 RECORDS

The following records requirements apply:

a. Pressure vessel owners and users shall maintain pressure vessel records. The preferred method of record keeping is to maintain data by individual vessel. Where vessels are grouped into classes, data may be maintained by vessel class. When inspections, repairs, or alterations are made on an individual vessel, specific data shall be recorded for that vessel.

b. Examples of information that may be maintained are vessel identification numbers; safety relief device information; and the forms on which results of inspections, repairs, alterations, or reratings are to be recorded. Any appropriate forms may be used to record these results. A sample pressure vessel inspection record is shown in Appendix C. A sample alteration or rerating of pressure vessel form is shown in Appendix D. Information on maintenance activities and events that affect vessel integrity should be included in the vessel records.

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APPENDIX A—ASME CODE EXEMPTIONS

The following classes of containers and pressure vessels are excluded from the specific requirements of this inspection code:

a. Pressure vessels on movable structures covered by jurisdictional regulations:

1. Cargo or volume tanks for trucks, ships, and barges.
2. Air receivers associated with braking systems of mobile equipment.
3. Pressure vessels installed in ocean-going ships, barges, and floating craft.

b. All classes of containers listed for exemption from the scope of Section VIII, Division 1, of the ASME Code:

1. Those classes of containers within the scope of other sections of the ASME Code other than Section VIII.
2. Fired process tubular heaters.
3. Pressure containers that are integral parts or components of rotating or reciprocating mechanical devices, such as pumps, compressors, turbines, generators, engines, and hydraulic or pneumatic cylinders where the primary design considerations or stresses are derived from the functional requirements of the device.
4. Any structure whose primary function is transporting fluids from one location to another within a system of which it is an integral part (that is, piping systems).
5. Piping components such as pipe, flanges, bolting, gaskets, valves, expansion joints, fittings, and the pressure-containing parts of other components such as strainers and devices which serve such purposes as mixing, separating, snubbing, distributing, and metering or controlling flow as long as the pressure-containing parts of these components are generally recognized as piping components or accessories.
6. A vessel for containing water under pressure, including vessels containing air, the compression of which serves only as a cushion, when the following limitations are not exceeded:
   (a) A design pressure of 300 pounds per square inch (2067.7 kilopascals).
   (b) A design temperature of 210°F (99°C).
7. A hot water supply storage tank heated by steam or any other indirect means when the following limitations are not exceeded:
   (a) A heat input of 200,000 British thermal units (211 × 108 joules) per hour.
   (b) A water temperature of 210°F (99°C).
   (c) A nominal water-containing capacity of 120 gallons (455 liters).
8. Vessels with an internal or external operating pressure not exceeding 15 pounds per square inch (103.4 kilopascals) but with no limitation on size.
9. Vessels with an inside diameter, width, height, or cross-section diagonal not exceeding 6 inches (15 centimeters) but with no limitation on their length or pressure.
10. Pressure vessels for human occupancy.

c. Pressure vessels that do not exceed the following volumes and pressures:

1. Five cubic feet (0.141 cubic meters) in volume and 250 pounds per square inch (1723.1 kilopascals) design pressure.
2. One-and-one-half cubic feet (0.042 cubic meters) in volume and 600 pounds per square inch (4136.9 kilopascals) design pressure.

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APPENDIX B—AUTHORIZED PRESSURE VESSEL INSPECTOR CERTIFICATION

B.1 Examination

To qualify an inspector within the scope of API 510, a written examination shall be administered to the inspector by a third party acceptable to API and the jurisdiction. This examination shall be based on the content of the latest edition of API 510 and the applicable portions of Sections V, VIII, and IX of the latest edition of the ASME Code.

B.2 Certification

B.2.1 An API certificate will be issued when an applicant is in compliance with API 510; however, if an applicant possesses an owner-user certification or commission for inspection of pressure vessels (as indicated below), it will be recognized as long as the applicant meets the experience and education requirements of API 510. An API 510 inspector certification may be issued when proof of one of the following is presented:

a. A jurisdictional agency certification. The applicant must have qualified by written examination under the laws, rules, and regulations of a jurisdiction.
b. A National Board of Boiler and Pressure Vessel Inspectors commission or owner-user commission. The applicant must be qualified by written examination under the rules and regulations of the National Board.

B.2.2 Subject to the approval of the appropriate jurisdiction during the first twelve months after the original June 1989 date of this appendix, owner-user inspectors who were employed full time as pressure equipment inspectors in the industry were eligible for a special one-time certification. Documentation of satisfactory performance on the job for the previous five years was required.

B.2.3 An API certificate for an authorized pressure vessel inspector is valid for three years from its date of issuance.

B.2.4 An API inspector certification is valid in all 50 states and in any other location that accepts API 510.

B.3 Retroactivity

Except as provided in B.2.1, the certification requirements of API 510 shall not be retroactive or interpreted as applying before the June 1989 date.

B.4 Recertification

B.4.1 Recertification by written test will be required for authorized pressure vessel inspectors who have not been actively engaged as authorized pressure vessel inspectors within the previous three years.

B.4.2 Recertification tests will be in accordance with all of the provisions contained in API 510.

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APPENDIX C—SAMPLE PRESSURE VESSEL INSPECTION RECORD

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STD.API/PETRO STD 510-ENGL 1997 ■ 0732290 0567588 461 ■

SAMPLE PRESSURE VESSEL INSPECTION RECORD

Form Date _____
Form No. _____
Owner or User _____
Vessel Name _____

Description

Name of Process _____
Location _____
Internal Diameter _____
Tangent Length/Height _____
Shell Material Specification _____
Head Material Specification _____
Internal Materials _____
Nominal Shell Thickness _____
Nominal Head Thickness _____
Design Temperature _____
Maximum Allowable Working Pressure _____
Maximum Tested Pressure _____
Design Pressure _____
Relief Valve Set Pressure _____
Contents _____
Special Conditions _____

Owner or User Number _____
Jurisdiction/National Board Number _____
Manufacturer _____
Manufacturer's Serial No. _____
Date of Manufacture _____
Contractor _____
Drawing Numbers _____
Construction Code _____
Joint Efficiency _____
Type Heads _____
Type Joint _____
Flange Class _____
Coupling Class _____
Number of Manways _____
Weight _____

Thickness Measurements

Sketch or Location Description	Location Number	Original Thickness	Required Minimum Thickness	Date

Comments (See Note 2) _____
Method _____
Authorized Inspector _____

Notes:
1. Use additional sheets, as necessary.
2. The location that each comment relates to must be described.

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APPENDIX D—SAMPLE REPAIR, ALTERATION, OR RERATING OF PRESSURE VESSEL FORM

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STD.API/PETRO STD 510-ENGL 1997 ■ 0732290 0567590 01T ■

SAMPLE REPAIR ALTERATION OR RERATING OF PRESSURE VESSEL FORM

Form Date _____
Form No. _____
Owner or User Name _____
Vessel Name _____

1. Original Vessel Identification Number _____
2. Original Vessel Location _____
3. Manufacturer _____ Serial No. _____
4. See attachments for additional data? ❑ Yes ❑ No
5. Original Construction Code _____
6. Original Maximum Allowable Working Pressure _____ Year Built _____
7. Original Design Temperature _____ Year Built _____
8. Original Minimum Design Metal Temperature _____ At Pressure _____
9. Original Test Pressure _____ Fluid _____ Position _____
10. Shell Material _____ Head Material _____
11. Shell Thickness _____ Head Thickness _____
12. Original Joint Efficiency _____
13. Original Radiography ❑ Yes ❑ No
14. Original PWHT ❑ Yes ❑ No
    If yes, _____ Temp (°F) _____ Time (Hrs)
15. Original Corrosion Allowance _____
16. Work on Vessel Classified as: ❑ Repair ❑ Alteration ❑ Rerating
17. Organization Performing Work _____
18. Construction Code for Present Work _____
19. New Vessel Identification Number (if Applicable) _____
20. New Vessel Location (if Applicable) _____
21. New Maximum Allowable Working Pressure _____
22. New Design Temperature _____
23. New Minimum Design Metal Temperature _____ At Pressure _____
24. New PWHT ❑ Yes ❑ No
    _____ Temp (°F) _____ Time (Hrs)
25. New Joint Efficiency, if Applicable E = _____
26. Type of Examination or Inspection Performed:
    ❑ radiographic ❑ ultrasonic
    ❑ magnetic particle ❑ penetrant
    ❑ visual ❑ other
27. New Pressure Test if Yes, Pressure _____ Test Medium _____ Test Position _____
28. New Corrosion Allowance _____
29. Describe work performed (attach drawings, calculations, and other pertinent data):

Statement of Compliance

We certify that the statements made in this report are correct and that all material and construction for and workmanship of this ❑ repair ❑ alteration, ❑ rerating conform to the requirements of the _____ Edition of API 510, *Pressure Vessel Inspection Code.*

_____
(repair, alteration, or rerating organization)

Signed _____
(authorized representative)

Date _____

Statement of Inspection

I, the undersigned, an inspector employed by _____, having inspected the work described above, state that to the best of my knowledge, the work has been satisfactorily completed in accordance with the _____ Edition of API 510, *Pressure Vessel Inspection Code.*

Signed _____

API 510 Certification Number _____

Date _____

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APPENDIX E—TECHNICAL INQUIRIES

E.1 Introduction

API will consider written requests for interpretations of API 510. API staff will make such interpretations in writing after consultation, if necessary, with the appropriate committee officers and the committee membership. The API committee responsible for maintaining API 510 meets regularly to consider written requests for interpretations and revisions and to develop new criteria as dictated by technological development. The committee's activities in this regard are limited strictly to interpretations of the standard or to the consideration of revisions to the present standard on the basis of new data or technology. As a matter of policy, API does not approve, certify, rate, or endorse any item, construction, proprietary device, or activity; thus, accordingly, inquiries requiring such consideration will be returned. Moreover, API does not act as a consultant on specific engineering problems or on the general understanding or application of the rules. If, based on the inquiry information submitted, it is the opinion of the committee that the inquirer should seek assistance, the inquiry will be returned with the recommendation that such assistance be obtained.

All inquiries that cannot be understood because they lack information will be returned.

E.2 Inquiry Format

Inquiries shall be limited strictly to requests for interpretation of the standard or to the consideration of revisions to the standard on the basis of new data or technology. Inquiries shall be submitted in the following format:

a. Scope. The inquiry shall involve a single subject or closely related subjects. An inquiry letter concerning unrelated subjects will be returned.

b. Background. The inquiry letter shall state the purpose of the inquiry, which shall be either to obtain an interpretation of the standard or to propose consideration of a revision to the standard. The letter shall provide concisely the information needed for complete understanding of the inquiry (with sketches, as necessary). This information shall include reference to the applicable edition, revision, paragraphs, figures, and tables.

c. Inquiry. The inquiry shall be stated in a condensed and precise question format. Superfluous background information shall be omitted from the inquiry, and where appropriate, the inquiry shall be composed so that "yes" or "no" (perhaps with provisos) would be a suitable reply. This inquiry statement should be technically and editorially correct. The inquirer shall state what he believes the standard requires. If in his opinion a revision to the standard is needed, he shall provide recommended wording.

The inquiry should be typed; however, legible handwritten inquiries will be considered. The name and the mailing address of the inquirer must be included with the proposal. The proposal shall be submitted to the following address: Director, Manufacturing, Distribution and Marketing Department, American Petroleum Institute, 1220 L Street, N.W., Washington, D.C. 20005-4070.

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